SMD SEMICONDUCTOR
Sarawak Microelectronics Design

Centre for Technology Excellence SARAWAK

THE LAYOUT OF 50MHZ RING OSCILLATOR

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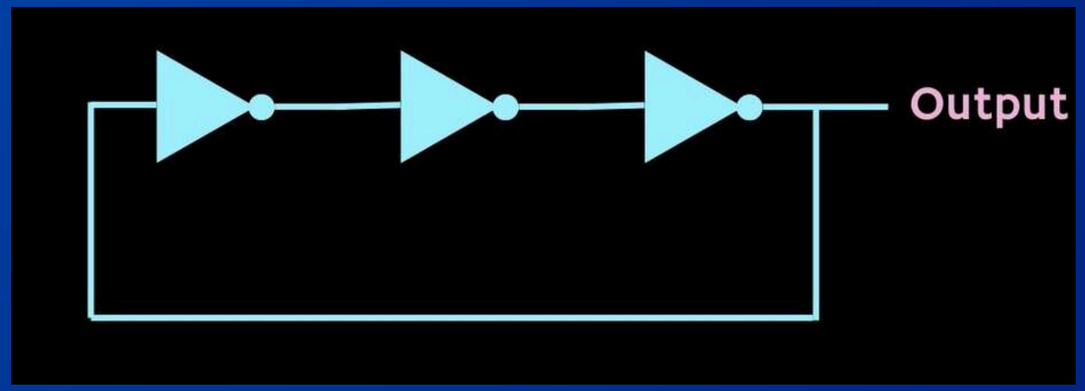
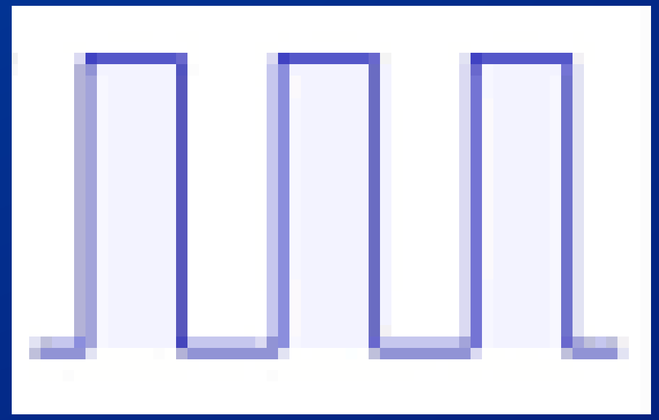
INTRODUCTION





- A ring oscillator is a simple, closed-loop circuit constructed by connecting an odd number of inverters (NOT gates) in a chain, with the output of the last inverter fed back into the first.
- Because of odd number of inversion, circuit is unstable(oscillating) between high and low voltage.
- This produces a continuous & self-sustaining oscillating clock signal

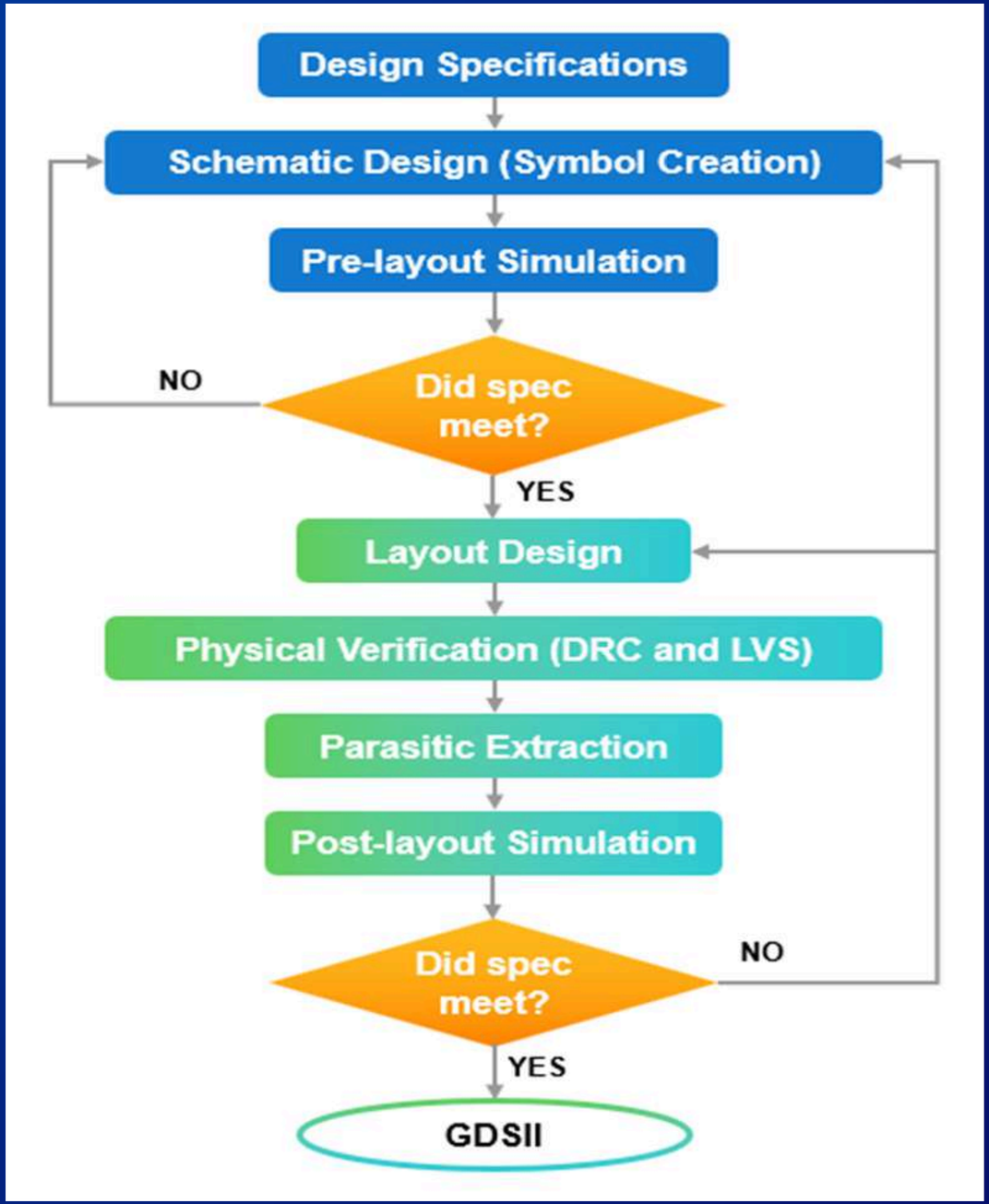
$$f = \frac{1}{2Nt_d}$$

- N is number of stage
- td is rime delay

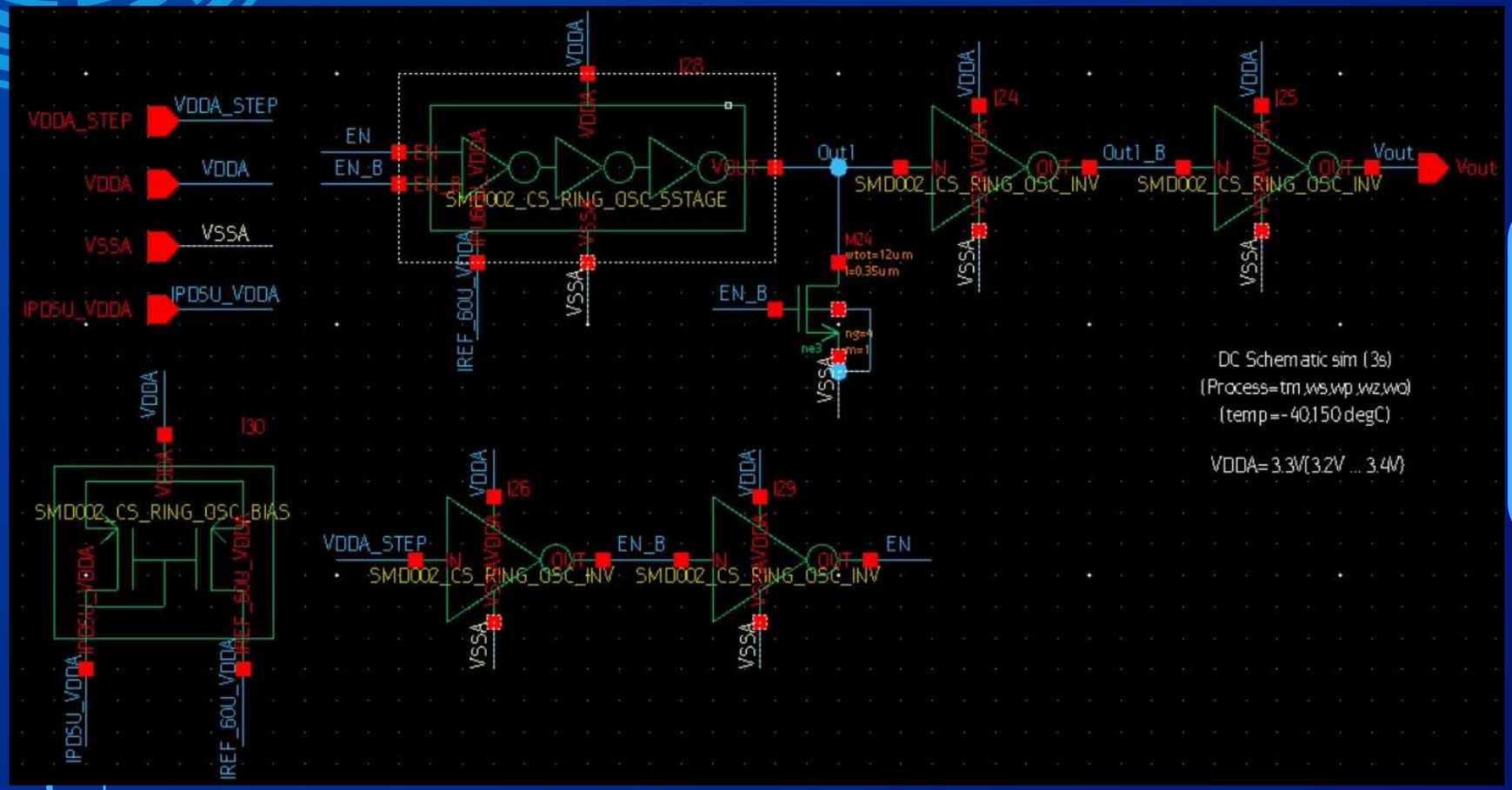
OBJECTIVE

1. Translate the ideal schematic of a 50MHz Ring Oscillator into a fully manufacturable physical layout design.
2. Implement advanced analog layout techniques(Common Centroid) placement and to ensure strict symmetry, balance and matched circuit.
3. Optimize floorplanning and metal routing to minimize parasitic capacitance (C) and resistance (R) ensuring the post-layout frequency remains accurately targeted at 50MHz.

LAYOUT DESIGN FLOW



SCHEMATIC OF RING OSCILLATOR



CONSIST OF:

1. Biasisng Circuit
2. 5 stage inverters
3. Inverters

THE SPECIFICATION REQUIRED

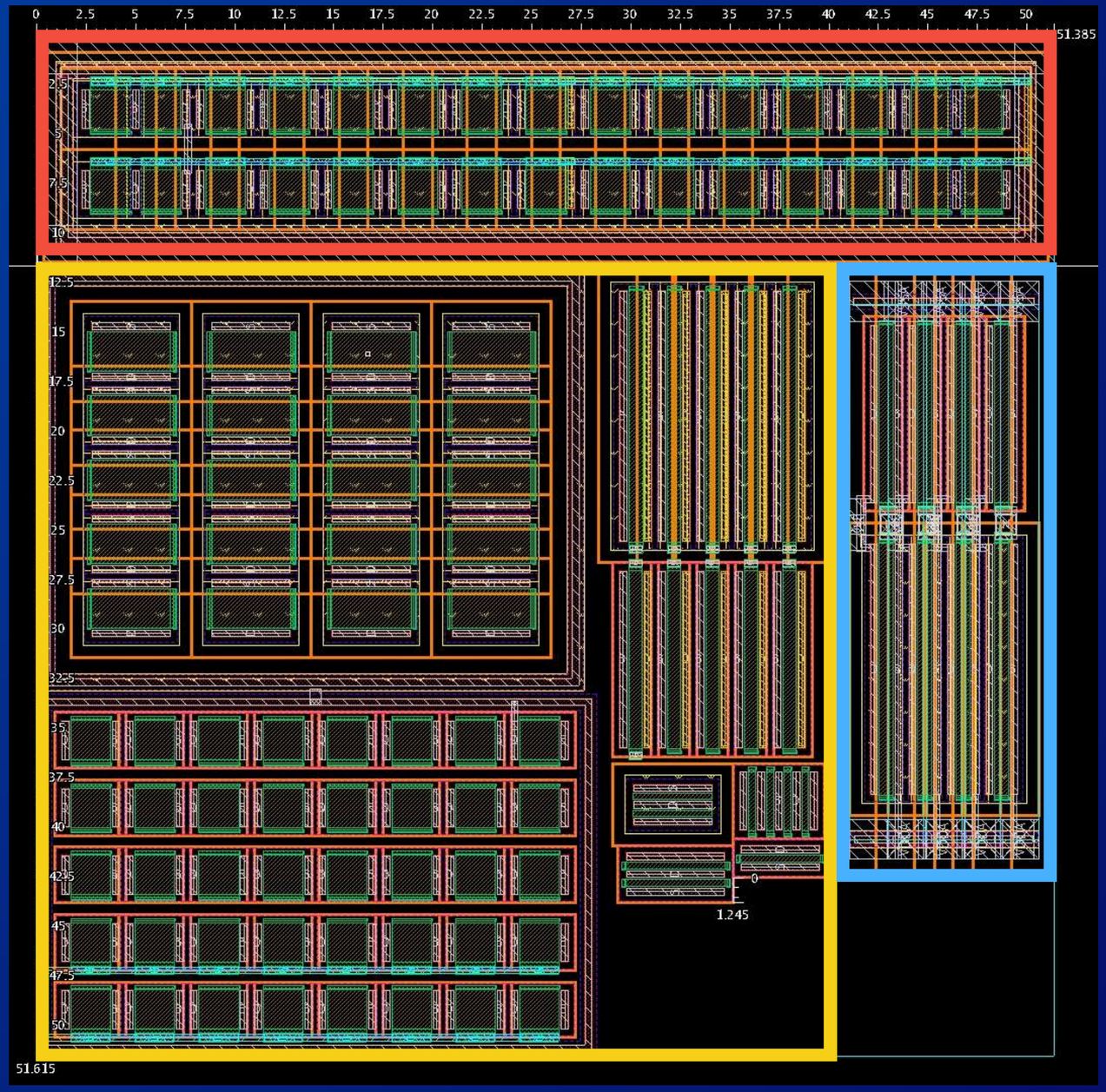
We require the specification of the Ring Oscillator as follows:

Parameters	Minimum	Typical	Maximum
Average Frequency (MHz)	45.689	50.7581	55.149
Average Duty Cycle (%)	50.426	51.5007	52.284
I_ON(uA)	426.17	479.6	542.47
I_OFF(nA)	1.9962	2.7691	7.9005

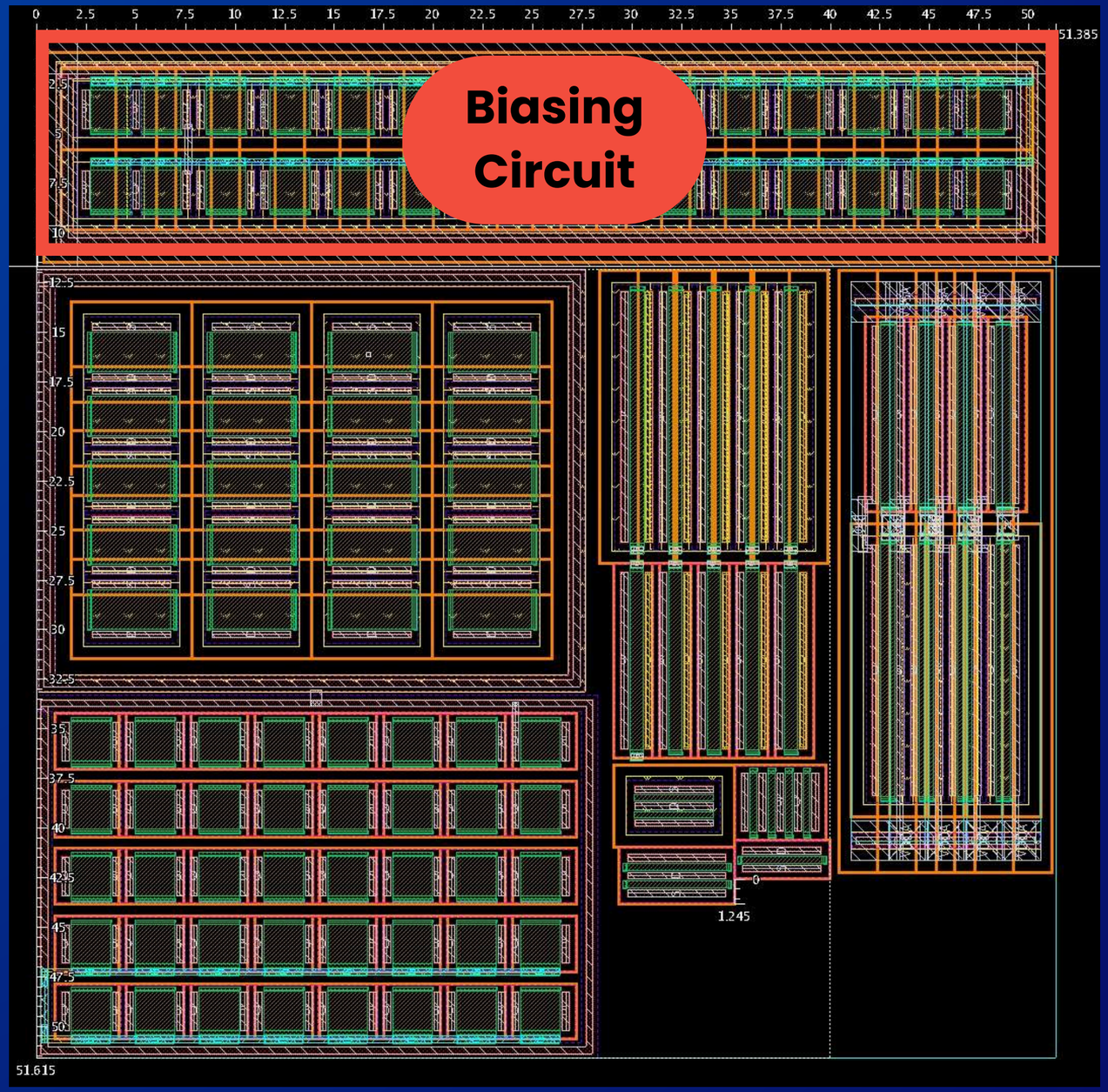
THE SPECIFICATION REQUIRED

- Average Frequency (MHz): This is the speed at which your oscillator generates a clock signal **(Ideal: 50 MHz)**
- Average Duty Cycle (%): This measures the percentage of time the clock signal is "HIGH" (logic 1) during one complete period. **(Ideal: 50%)**
- I_ON (uA): This is the dynamic or active current. It represents the power consumed by the ring oscillator while it is actively running and switching states. **(Typical: 479.6 uA)**
- I_OFF (nA): This is the static or leakage current. It is the tiny amount of current the circuit draws when it is powered on but not actively switching. **(Ideal: 0A)**

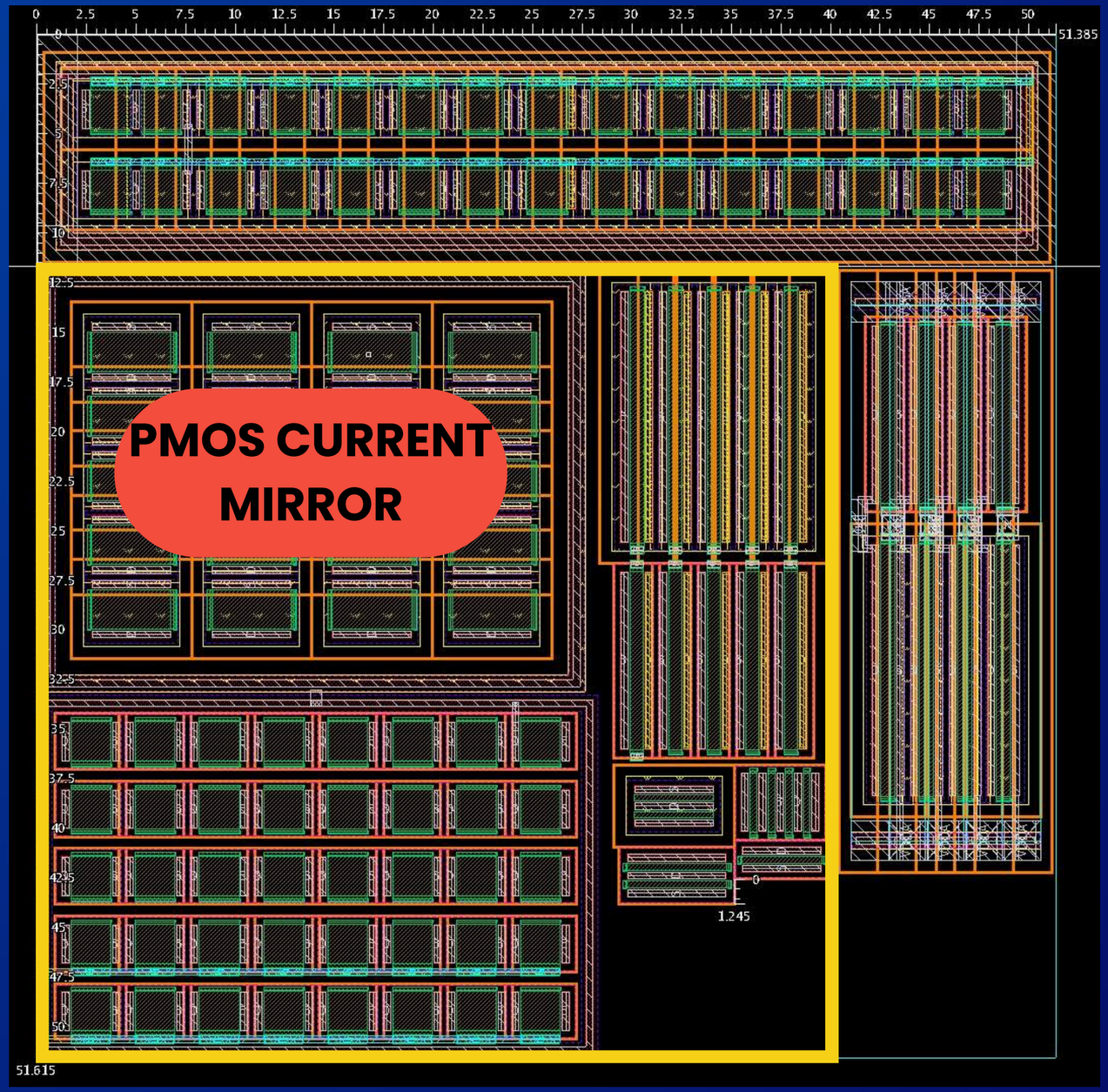
THE FLOORPLANNING OF LAYOUT



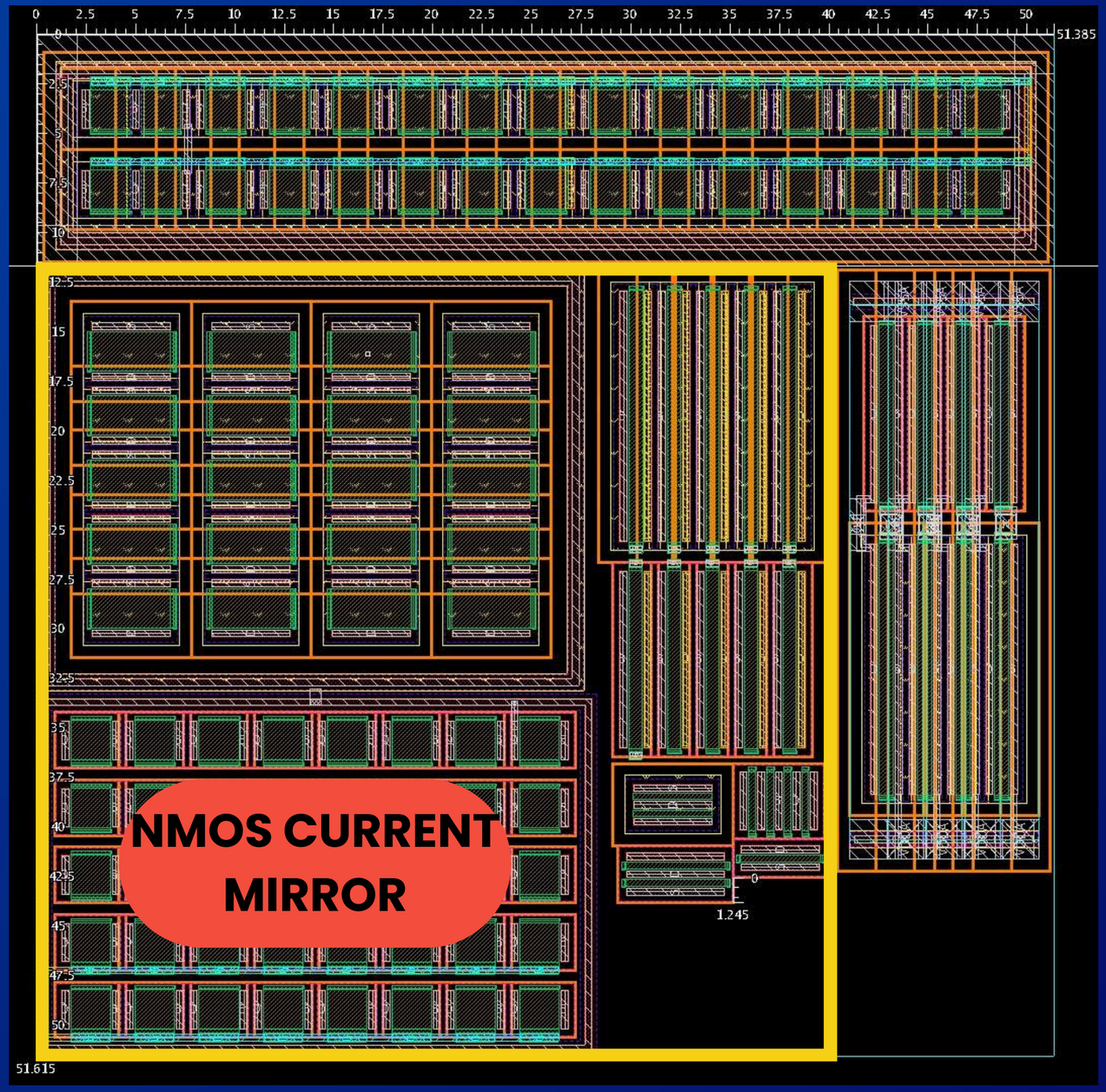
THE FLOORPLANNING OF LAYOUT



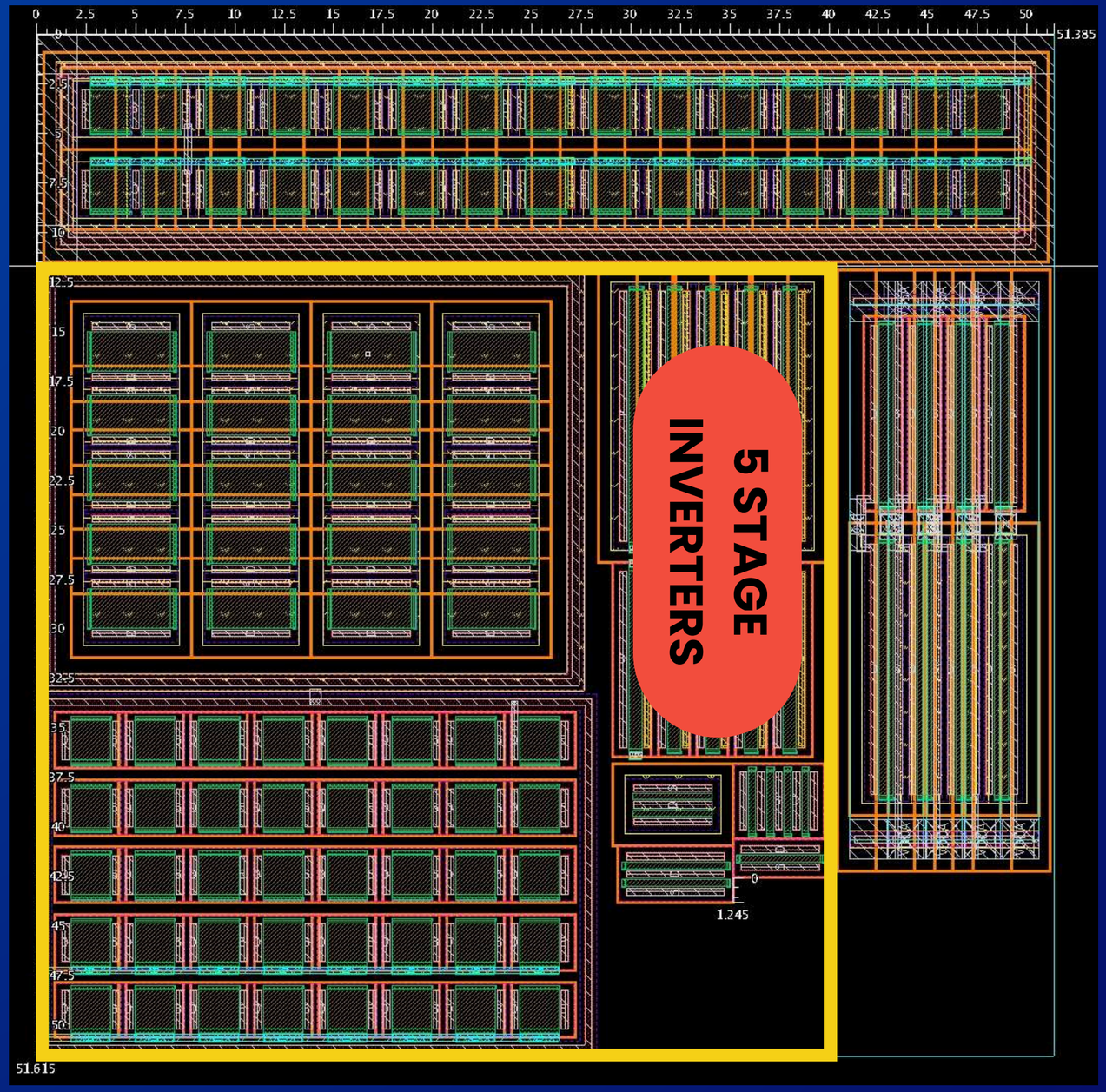
THE FLOORPLANNING OF LAYOUT



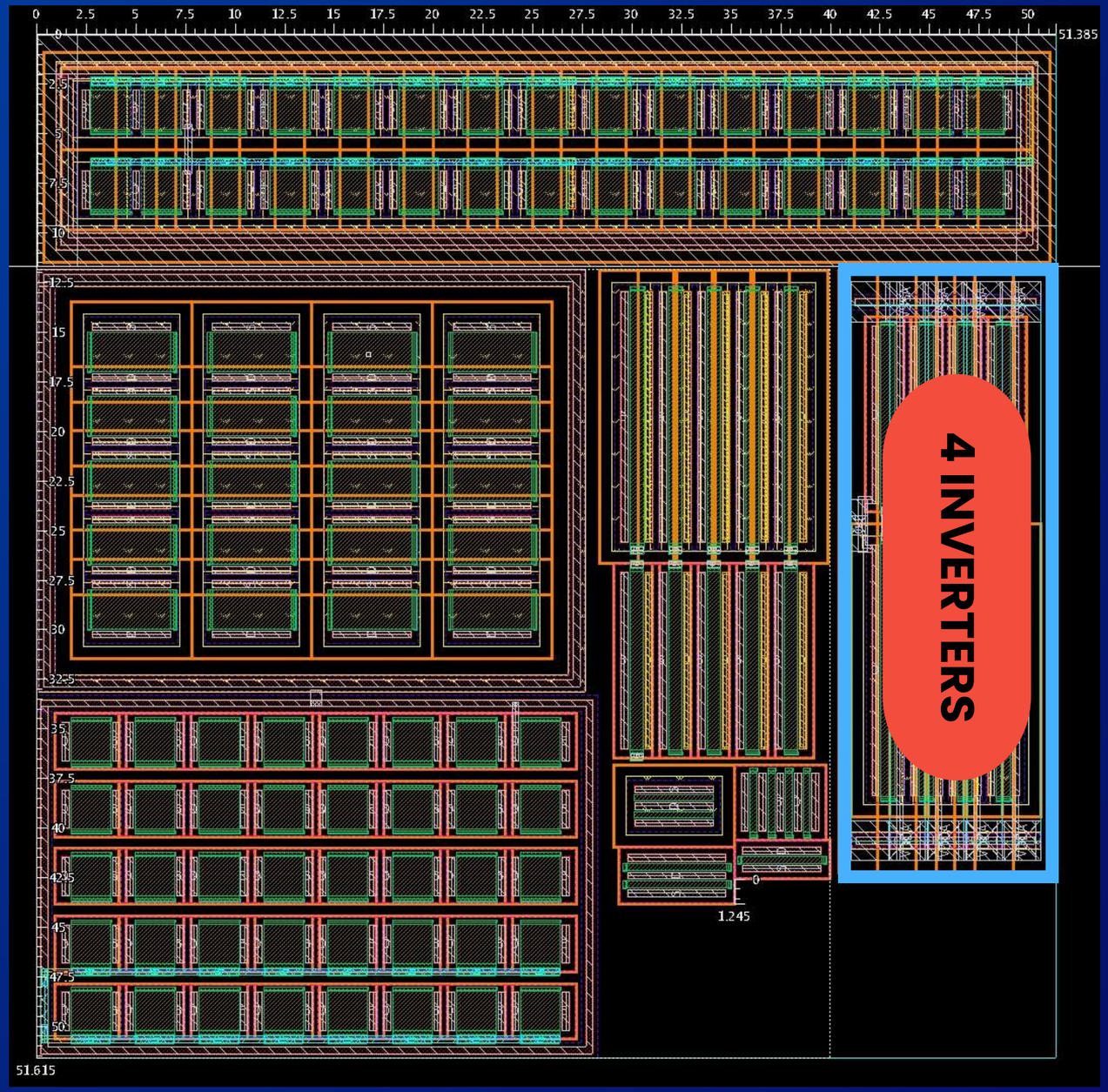
THE FLOORPLANNING OF LAYOUT



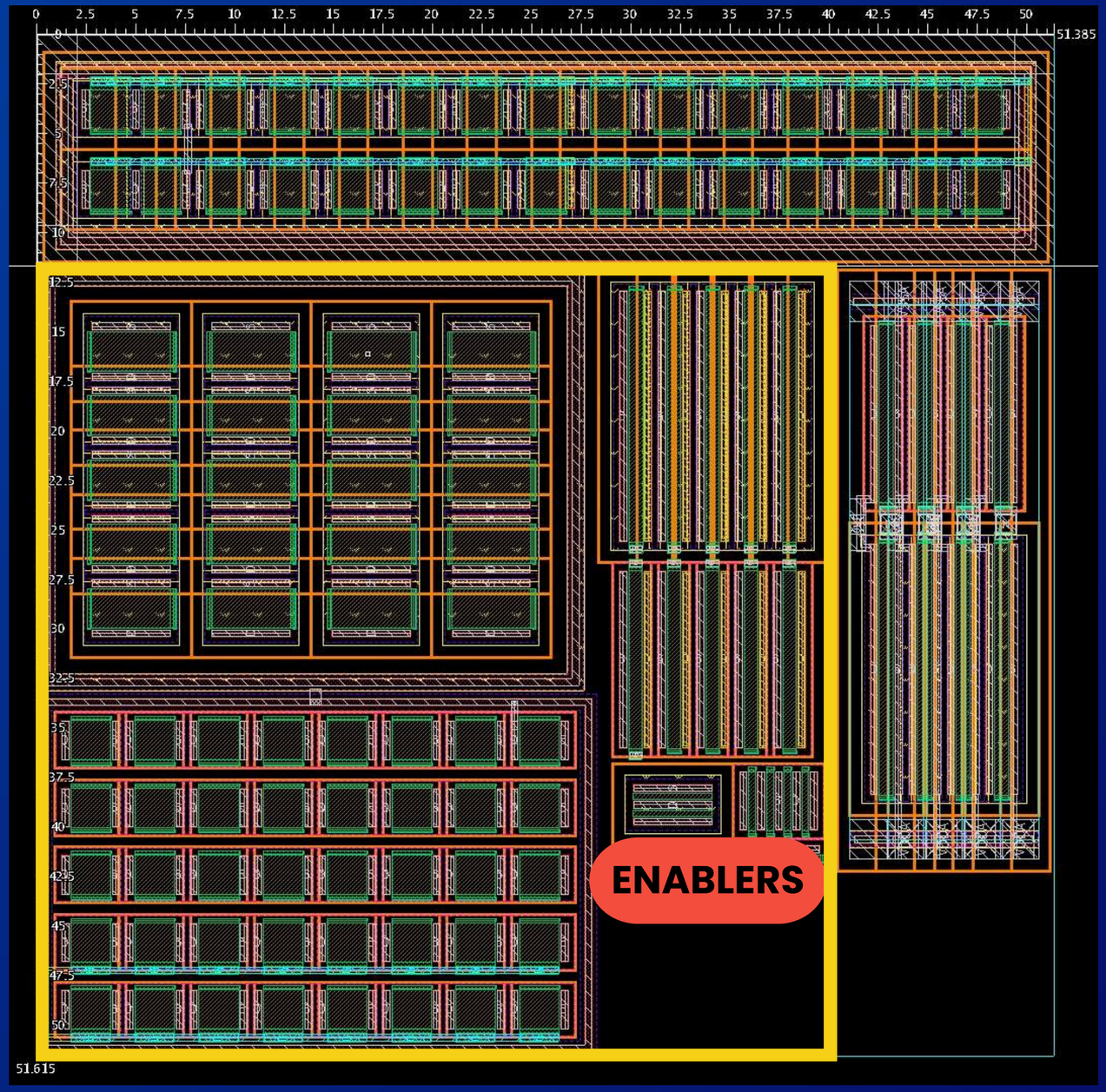
THE FLOORPLANNING OF LAYOUT



THE FLOORPLANNING OF LAYOUT



THE FLOORPLANNING OF LAYOUT

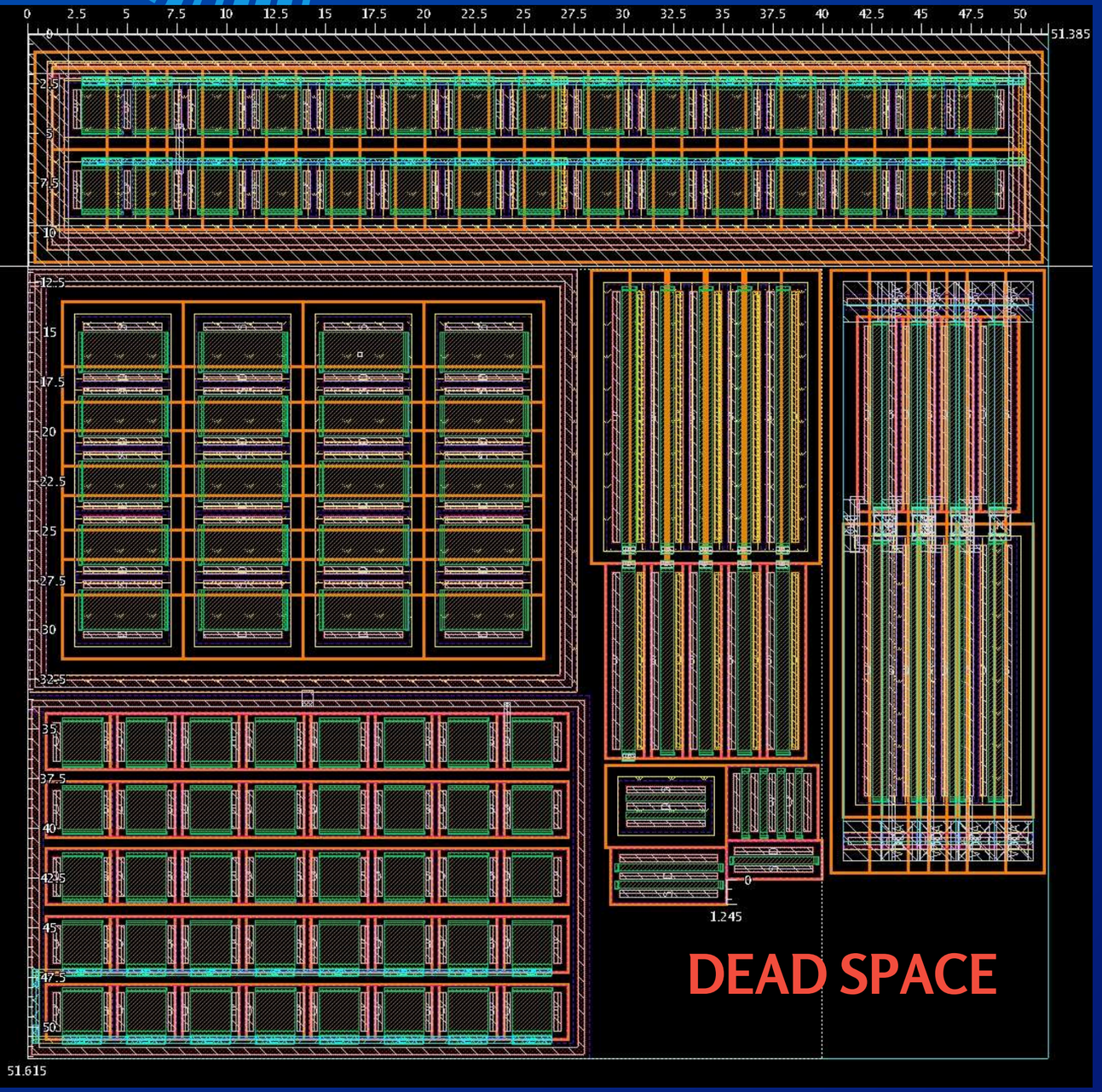


THE FLOORPLANNING STRATEGY

- Floorplanning is an early stage in the layout design flow which defines the chip's size, shape, functional block placements on a silicon die.

1. Implementation of modular placement - The blocks are group into similar transistor types (NMOS & PMOS).
2. Matching properties - Implementing Common Centroid and placing reference current at the middle.
3. Noise Isolation - Using Guard Ring around critical block to prevent latch-up and noise coupling.
4. Process Uniformity - Using dummy transistors to reduce non-uniformity during manufacturing processes(Lithography & Etching)

THE FLOORPLANNING OF STRATEGY – CONTD



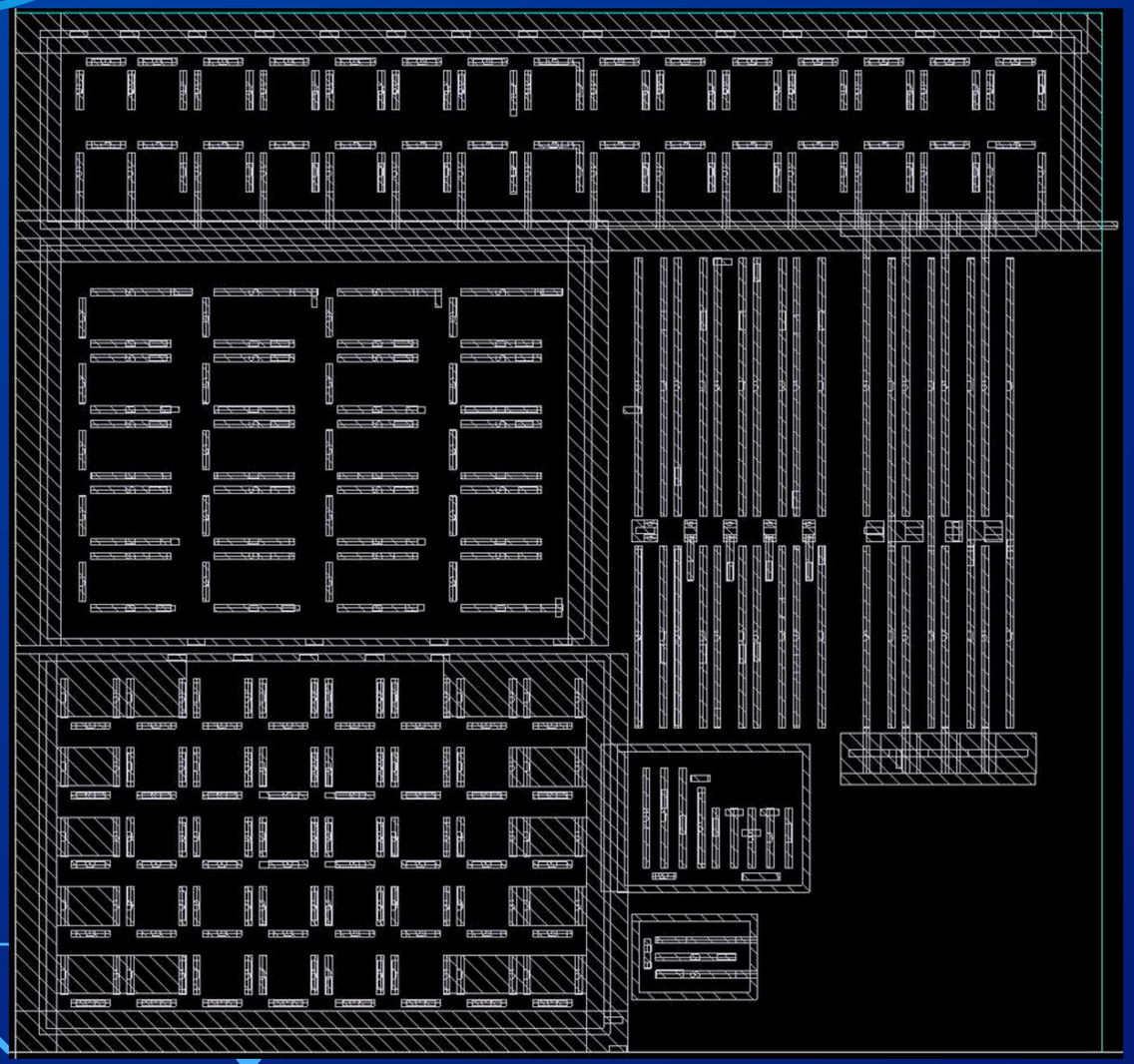
- Each blocks are placed nearer as possible such that the deadspace is minimized. (Minimum spacing rule)
- The available deadspace is spared for the purpose of routing/interconnection.
- The size of the layout is (51.385um x 51.615um) - Which is quite ‘squarish’.

ROUTING METHODOLOGY

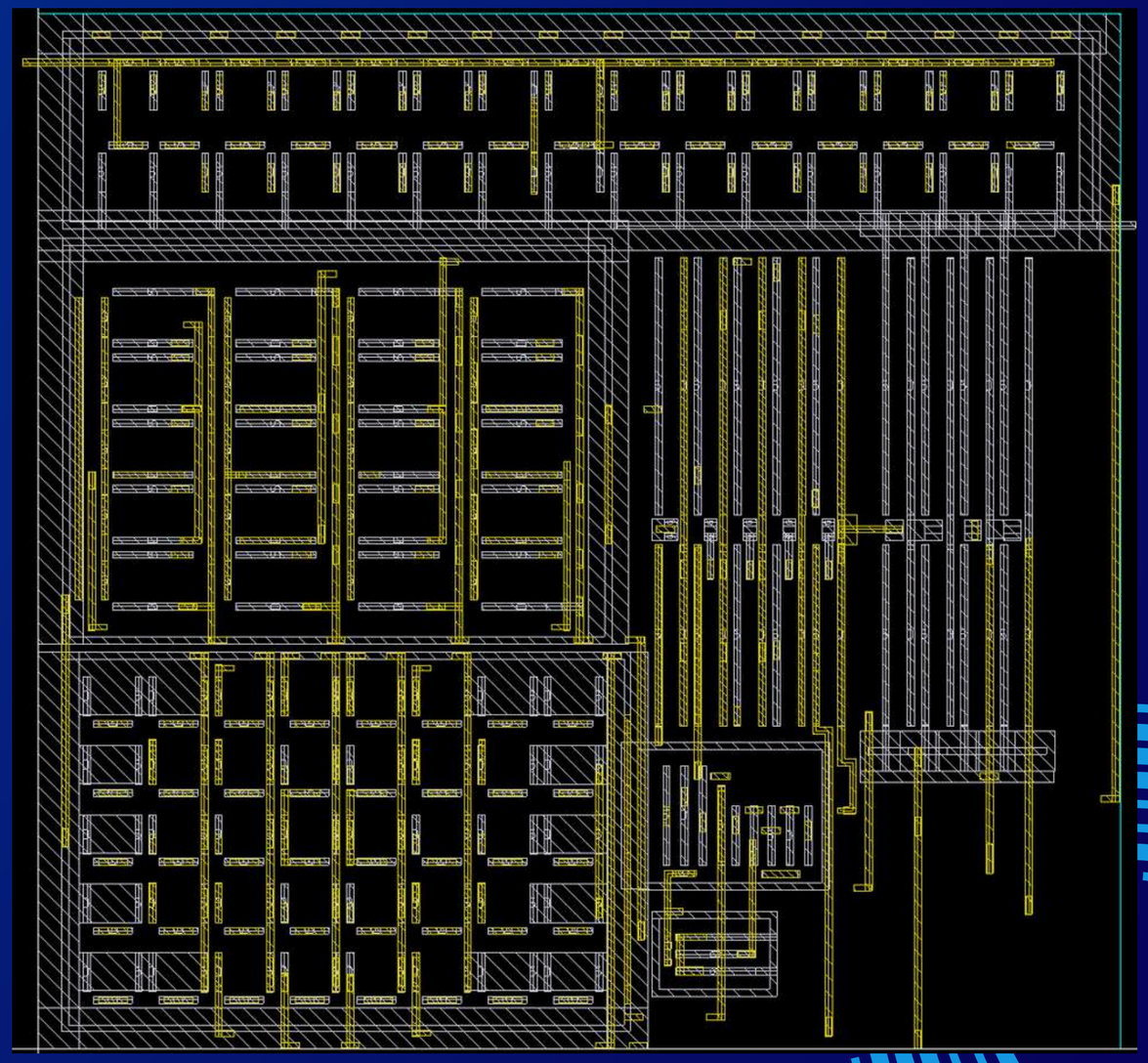
- Short-Path Feedback: Minimized the length of the wire connecting the last stage back to the first to reduce feedback latency.
- Symmetrical Interconnects: Ensured that the metal interconnectiong has identical width i.e. 0.34um for all metal.
- Metal Layer Strategy:
  - MET 1 is primarily used for internal transistor connection.
  - MET 2 used in vertical direction only.
  - MET 3 used in horizontal direction only.
- Doube Via to ensure reliability
- Power Strapping: Implemented thick metal rails of 2 um thickness for VDDA and VSSA to minimize IR drop and ensure stable oscillation.

ROUTING METHODOLOGY – METAL LAYER

METAL 1

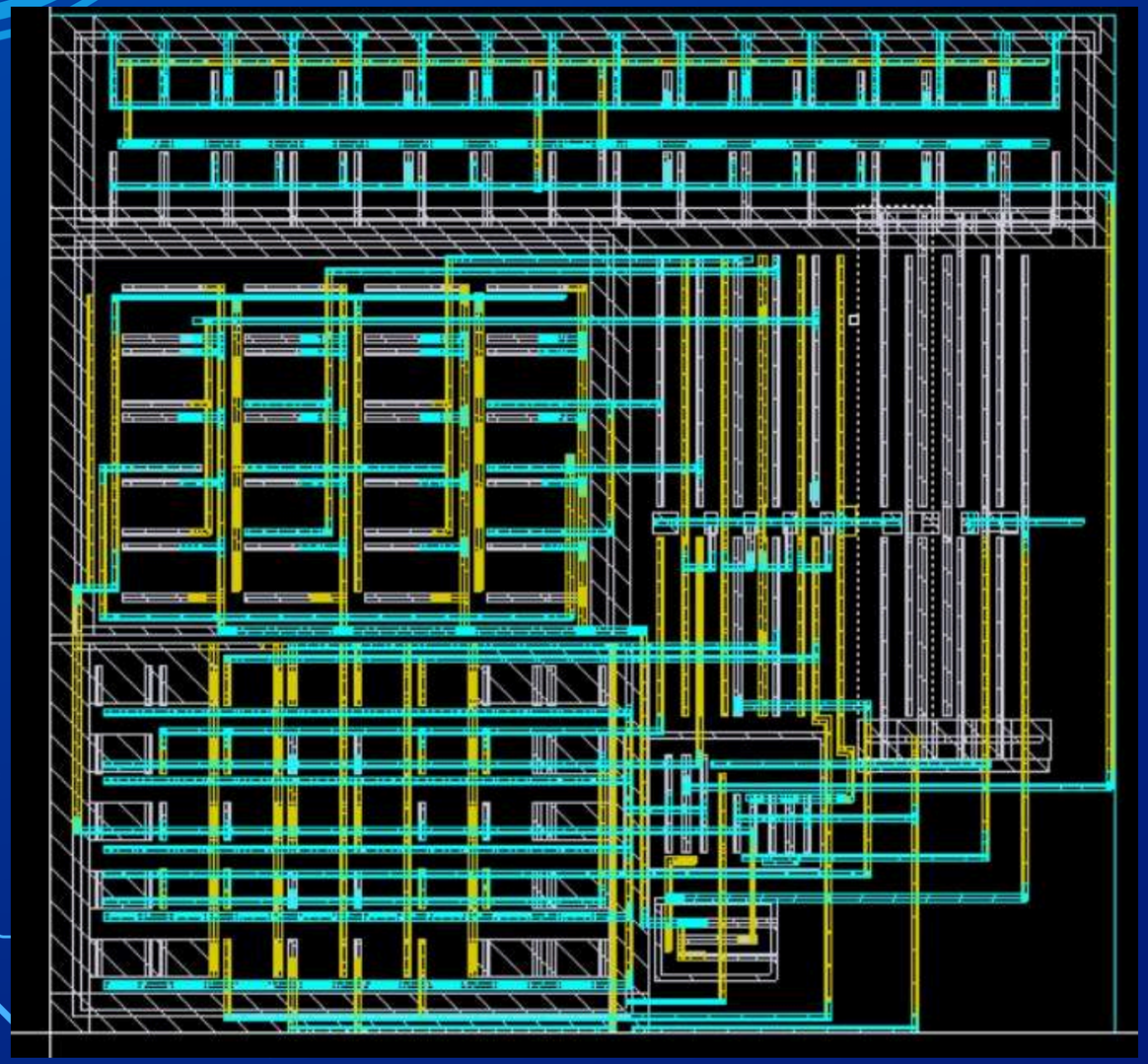


METAL 2

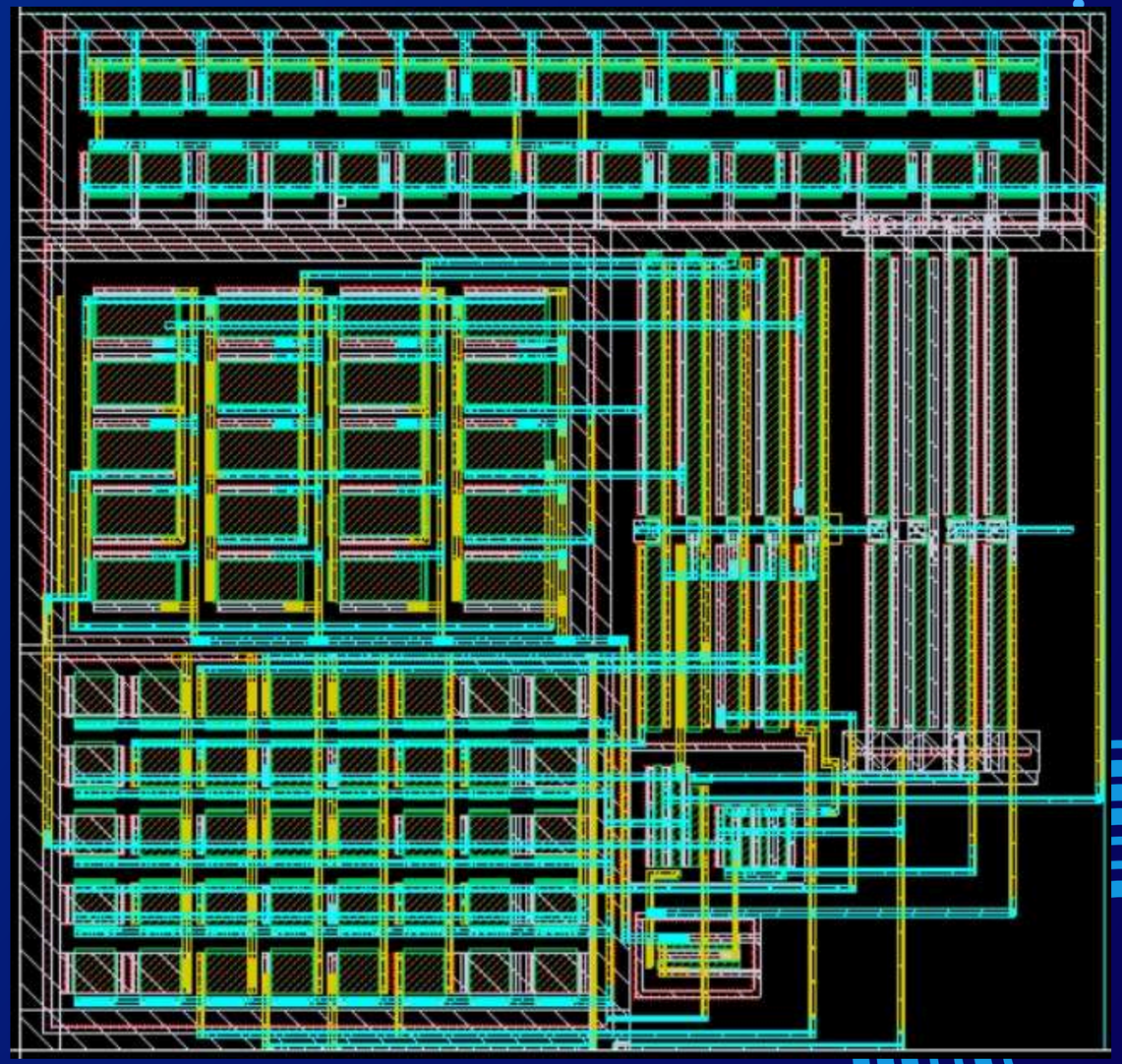


ROUTING METHODOLOGY – METAL LAYER

METAL 3

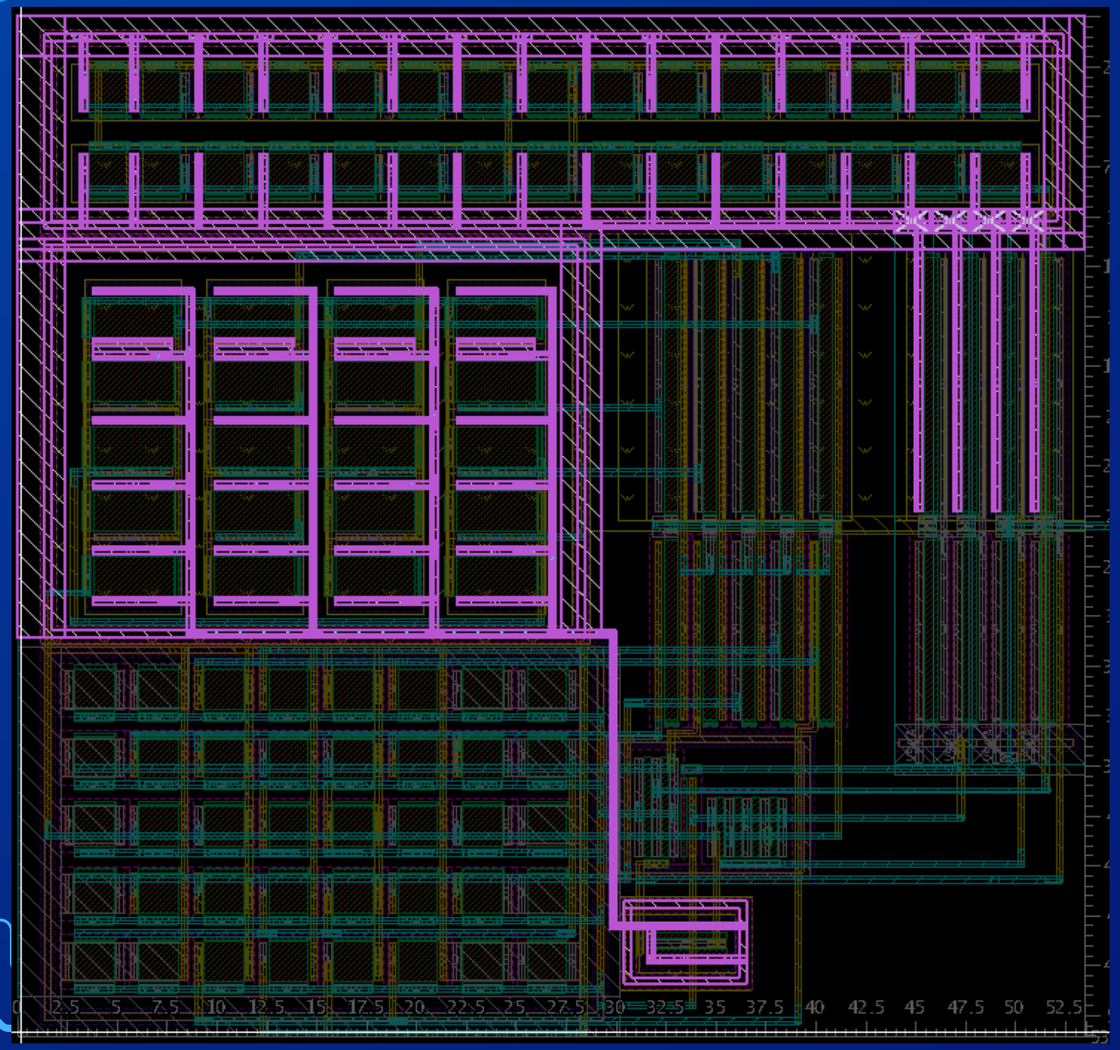


ROUTING LAYER

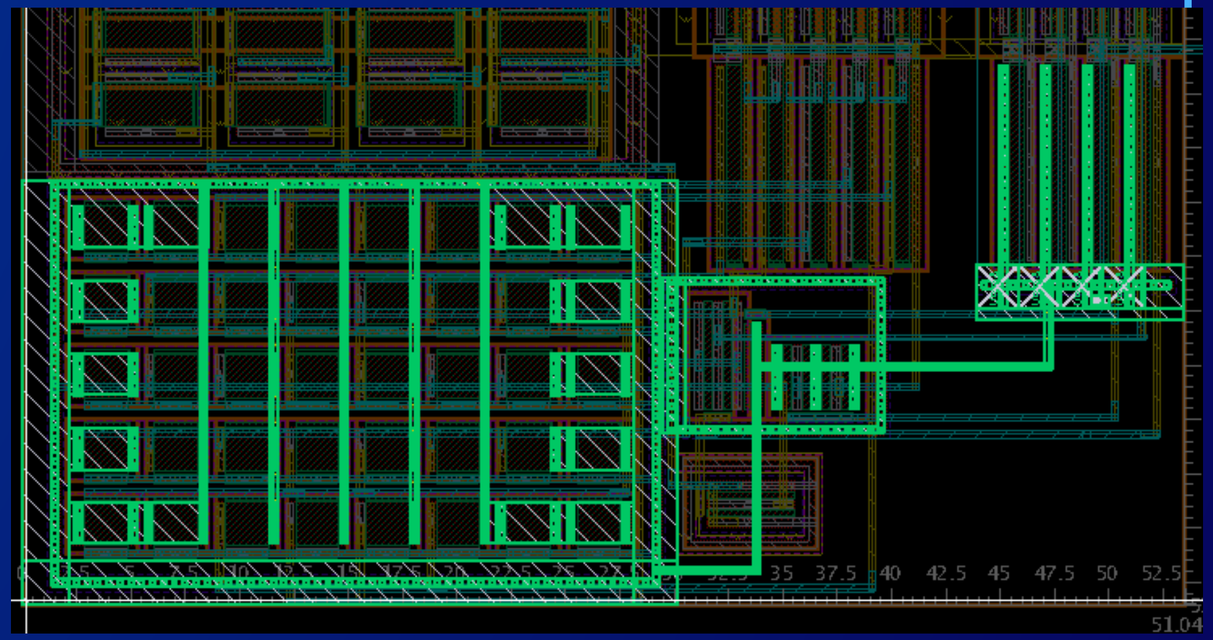


ROUTING METHODOLOGY – POWER LINE

VDDA

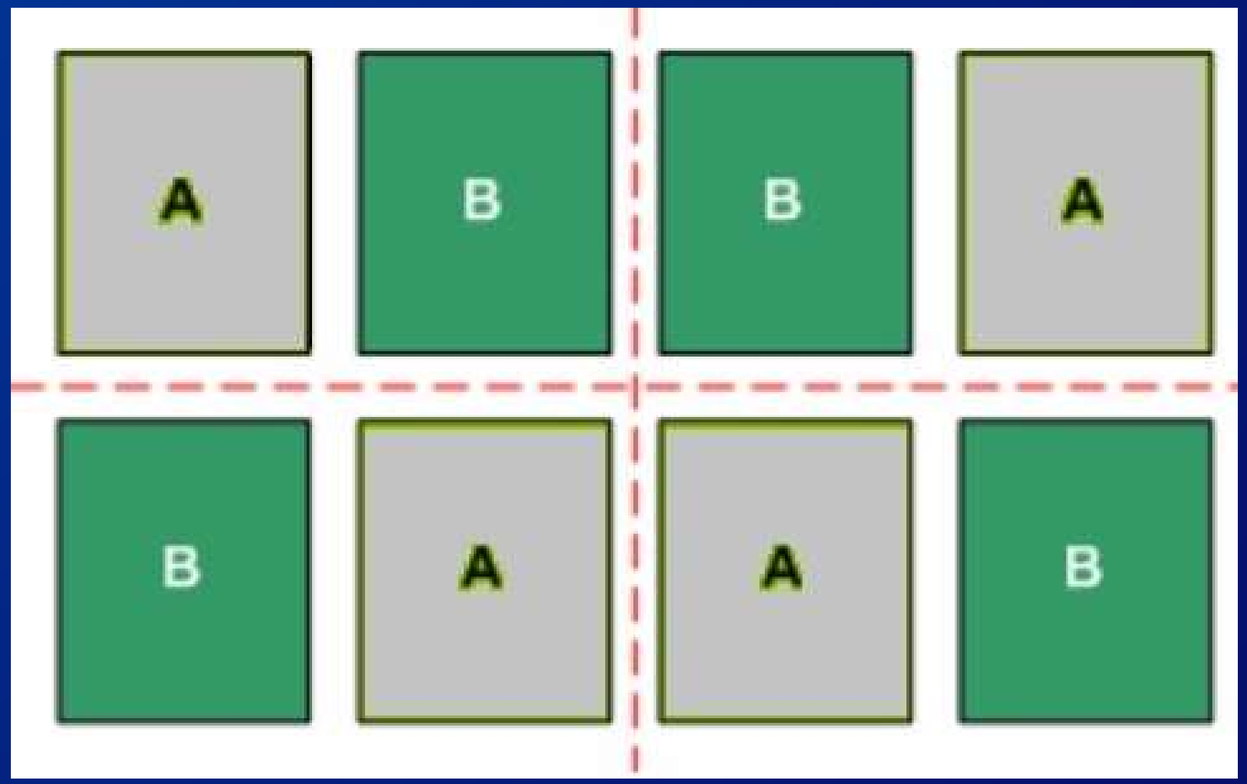


VSSA

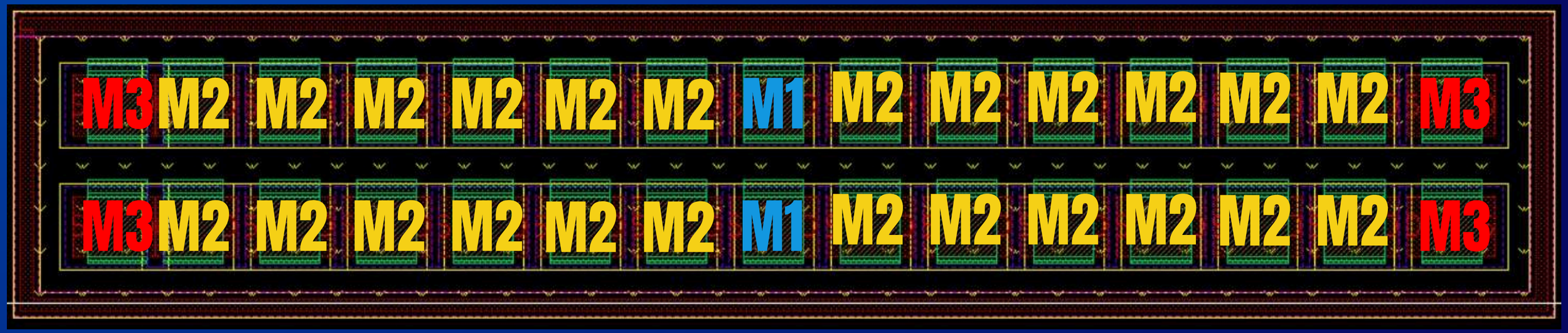


ROUTING METHODOLOGY – MATCHING

- A matching technique used to align the geometric center (centroid) of two or more matched devices. By interleaving or cross-coupling transistor fingers i.e. AB/BA pattern, the layout cancels out linear process gradients which would cause a mismatch between devices.

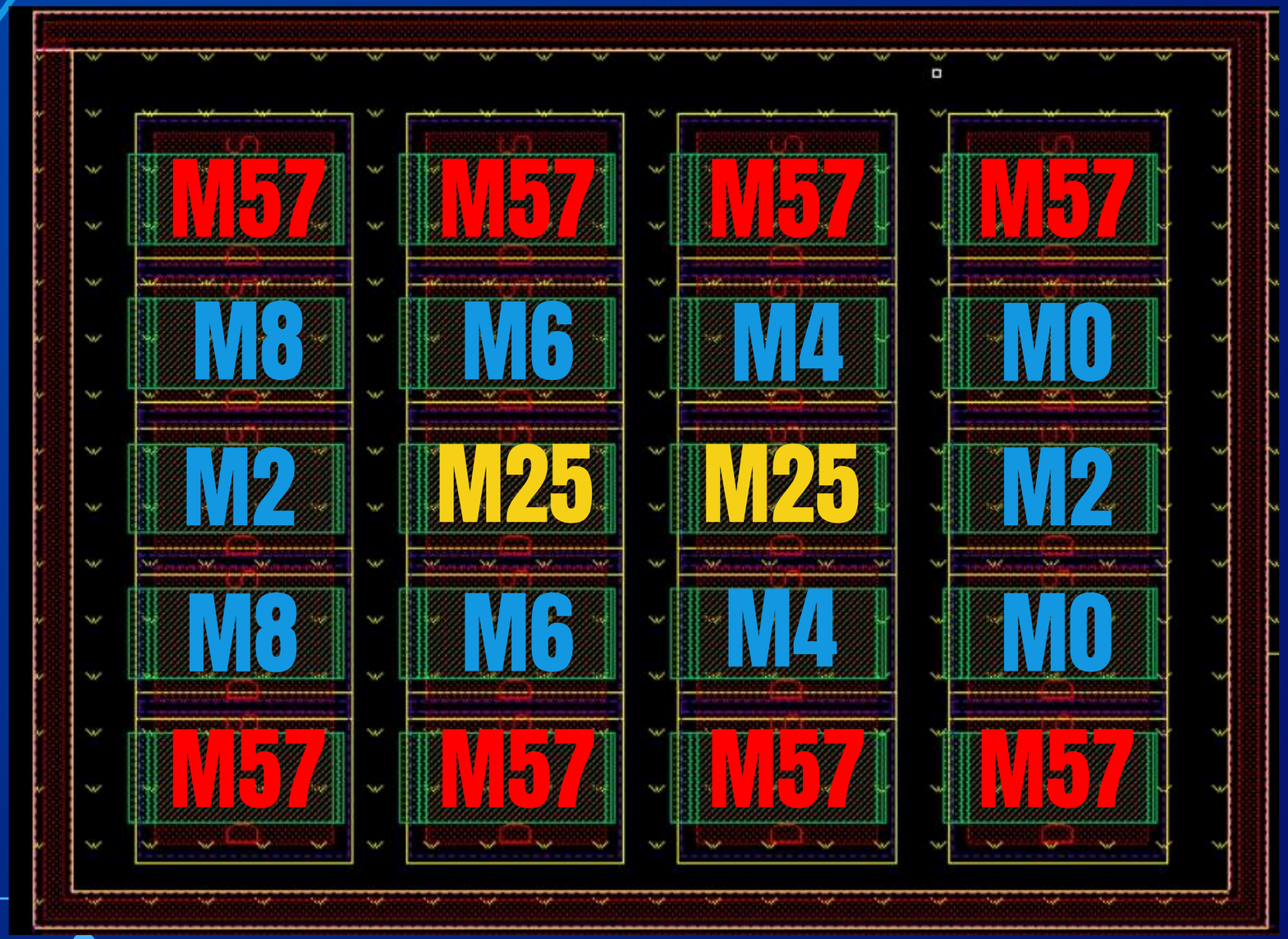


MATCHING BLOCK – BIASING CIRCUIT



THIS IS COMMON CENTROID!

MATCHING BLOCK – PMOS CURRENT MIRROR



THIS IS COMMON CENTROID!

MATCHING BLOCK – NMOS CURRENT MIRROR

THIS IS COMMON CENTROID!

DRC AND LVS CHECK

Design Rule Check

- DRC is the process of verifying that the geometric shapes in the physical layout conform to the strict manufacturing constraints (design rules) established by the semiconductor foundry.
- This includes the minimun spacing (in um) between metal, DIFF, contact spacing, etc.

- Min spacing of MET I: 0.23
- Min spacing of MET II: 0.28
- Min spacing of MET III: 0.28
- Min spacing of DIFF: 0.28

Layout Versus Schematic

- The verification process that extracts a netlist (components and their connections) directly from the layout drawn and compares it against the netlist of the original schematic.

DRC AND LVS CHECK

VUE: [/home/vjx/synopsys_custom/SMD002_CS_RING_OSC_50MHz.icv.drc/SMD002_CS_RING_OSC_50MHz.vue]

File View Tools Windows Help

Load Results × | Run Summary ×

LAYOUT ERRORS RESULTS

CLEAN

DRC Error Statistics

```
Library name:    Optimize_RO_II
Structure name:  SMD002_CS_RING_OSC_50MHz
Generated by:    IC Validator RHEL64 U-2022.12-SP5.9338971 2023/10/26
Runset name:     /home/vjx/synopsys_custom/SMD002_CS_RING_OSC_50MHz.icv.drc/xh018_1151_DRC.drc.r
User name:       vjx
Time started:    2026/03/10 11:57:07PM
Time ended:      2026/03/10 11:57:12PM

Called as: icv -f openaccess -i Optimize_RO_II -c SMD002_CS_RING_OSC_50MHz -oa_view layout -oa_l
```

VUE: [/home/vjx/synopsys_custom/SMD002_CS_RING_OSC_50MHz.icv.lvs/SMD002_CS_RING_OSC_50MHz.vue]

File View Tools Windows Help

Load Results × | Run Summary × | LVS Errors ×

LVS Compare Results: PASS

DRC and Extraction Results: CLEAN

[SMD002_CS_RING_OSC_50MHz, SMD002_CS_RING_OSC_50MHz]

Netlist Extraction Statistics

```
Library name:    Optimize_RO_II
Structure name:  SMD002_CS_RING_OSC_50MHz
Generated by:    IC Validator RHEL64 U-2022.12-SP5.9338971 2023/10/26
Runset name:     /home/vjx/synopsys_custom/SMD002_CS_RING_OSC_50MHz.icv.lvs/xh018_1151_StarRCXT.lvs.rs
User name:       vjx
Time started:    2026/03/10 11:57:58PM
Time ended:      2026/03/10 11:58:08PM

Called as: icv -f openaccess -i Optimize_RO_II -c SMD002_CS_RING_OSC_50MHz -oa_view layout -oa_lib_defs /home/vjx/work/lib.defs -oa
```

Layout vs. Schematic Statistics

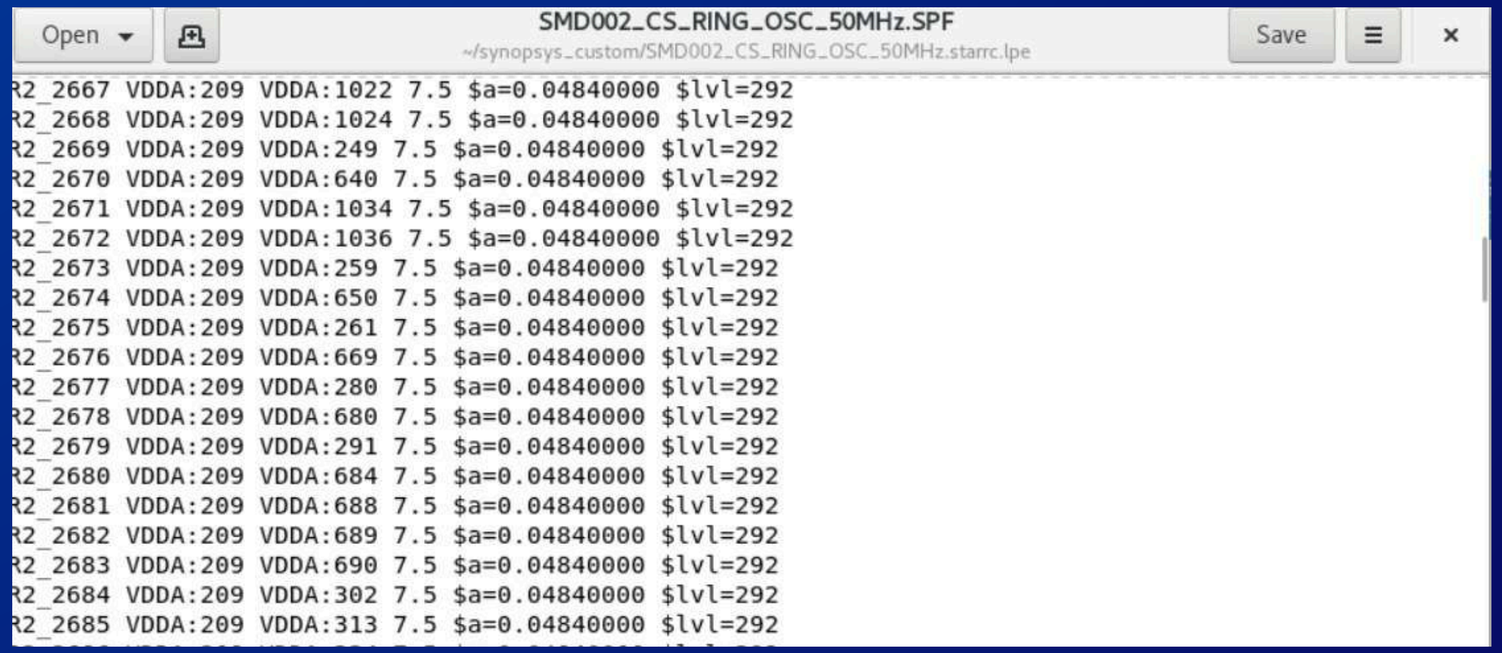
Schematic: /home/vjx/synopsys_custom/SMD002_CS_RING_OSC_50MHz.icv.lvs/SMD002_CS_RING_OSC_50MHz.sch_out.gz

LVS Errors:

```
blackbox cells checked: 0
  passed  0
  failed  0
equivalence points checked: 1
  passed  1
  failed  0
```

PARASITIC EXTRACTION

- Parasitic Extraction is the process of calculating the unintended resistance & capacitance(RC) that are created due to the physical geometry of the layout
- The presence of parasitic can cause the circuit to oscillate slowly and increase propagation delay.
- Parasitic extraction is done to take into account the unintented resistance(causing IR drop) and capacitance which leads to RC delay. The parasitic values are extracted into SPF file.



DETERMINING THE PARASITIC VALUES

- Using the LINUX command :

→ grep "VSSA" SMD002_CS_RING_OSC_50MHz.SPF | grep "^R" | awk '{sum += $4} END {printf "Total Resistance: %.2f Ohms\n", sum}'

→ grep "VSSA" SMD002_CS_RING_OSC_50MHz.SPF | grep "^C" | awk '{sum += $4} END {printf "Total Resistance: %.2f Ohms\n", sum}'

```
[vjx@smdapt102-almalinux SMD002_CS_RING_OSC_50MHz.starrc.lpe]$ grep "^R" SMD002_
CS_RING_OSC_50MHz.SPF | awk '{sum += $4} END {printf "Total Resistance: %.4f Ohm
s\n", sum}'
Total Resistance: 31840.5157 Ohms
[vjx@smdapt102-almalinux SMD002_CS_RING_OSC_50MHz.starrc.lpe]$
```

$$t_{d_{total}} = t_d + t_{RC} = t_d + RC$$

```
vjx@smdapt102-almalinux SMD002_CS_RING_OSC_50MHz.starrc.lpe]$ grep "^C" SMD002_
S_RING_OSC_50MHz.SPF | awk '{sum += $4} END {printf "Total Capacitance: %.18f F
n", sum}'
otal Capacitance: 0.000000000000410066 F
vjx@smdapt102-almalinux SMD002_CS_RING_OSC_50MHz.starrc.lpe]$
```

- Because of the parasitic values, it will introduce a delay so-called RC delay.
- This RC delay will be added up in our circuit, making our oscillator more sluggish.
- Additional of 13ns of delay.

PRE-LAYOUT SIMULATION (PVT ANALYSIS)

Corner	Corner Variable temp	Corner Variable VDDA	Corner Variable xh018.lib	Corner Variable param.lib	Corner Variable config.lib	avgfreq tran	avg_dutycycle tran	I_ON tran	I_OFF tranMeasure
Filter	< Filter	< Filter	Filter	< Filter	Filter	< Filter	< Filter	< Filter	< Filter
typ	25	3.3	tm			52.8218M	51.9168	474.112u	-2.7185n
corner1.1						56.564M	52.0024	471.478u	-2.1845n
corner1.2			wo			56.6386M	52.572	465.224u	-2.1742n
corner1.3		3.2	wp			55.2558M	52.12	497.057u	-1.968n
corner1.4			ws			57.3736M	51.7192	448.295u	-2.433n
corner1.5			wz			56.4657M	51.2036	477.052u	-2.1651n
corner1.6			tm			53.9235M	51.9918	489.332u	-2.178n
corner1.7			wo			54.0246M	52.6386	482.078u	-2.188n
corner1.8	-40	3.3	wp			52.5581M	52.0071	517.325u	-1.965n
corner1.9			ws			54.9182M	51.7766	463.947u	-2.4028n
corner1.10			wz			53.82M	51.5035	493.351u	-2.1689n
corner1.11			tm			51.4322M	52.2413	505.309u	-2.1729n
corner1.12			wo			51.5402M	52.5874	499.533u	-2.1838n
corner1.13		3.4	wp			50.0403M	52.25	534.652u	-1.9585n
corner1.14			ws			52.525M	51.884	479.882u	-2.3977n
corner1.15			wz			51.3334M	51.487	511.761u	-2.1632n
corner1.16			tm			55.1947M	51.7888	459.36u	-2.7281n
corner1.17			wo	3s	default	55.2557M	52.3506	453.949u	-2.7376n
corner1.18		3.2	wp			54.1393M	51.8092	484.453u	-2.4544n
corner1.19			ws			55.6439M	51.5816	436.232u	-3.0322n
corner1.20			wz			55.0899M	50.9738	464.091u	-2.7217n
corner1.21			tm			52.8218M	51.9168	474.112u	-2.7185n
corner1.22			wo			52.9145M	52.3432	469.572u	-2.7271n
corner1.23	25	3.3	wp			51.6802M	51.8342	501.441u	-2.4449n

corner1.24			ws			53.4819M	51.7821	449.865u	-3.0167n
corner1.25			wz			52.7201M	51.1491	479.418u	-2.7094n
corner1.26			tm			50.5669M	52.0438	488.837u	-2.705n
corner1.27			wo			50.6694M	52.5026	483.814u	-2.7181n
corner1.28		3.4	wp			49.3691M	51.9742	517.093u	-2.4407n
corner1.29			ws			51.3768M	51.867	463.941u	-2.9841n
corner1.30			wz			50.4713M	51.3199	494.048u	-2.7028n
corner1.31			tm			52.0862M	51.4062	445.345u	-5.1459n
corner1.32	150	3.2	wo			52.1216M	51.7242	442.786u	-5.8534n
corner1.33			wp			51.5719M	51.4936	470.149u	-7.7592n
corner1.34			ws			51.8412M	51.0508	422.498u	-4.9673n
corner1.35			wz			52.0162M	50.7744	448.436u	-4.8723n
corner1.36			tm			50.219M	51.5079	458.003u	-5.1373n
corner1.37			wo			50.2489M	51.8863	455.209u	-5.8524n
corner1.38	150	3.3	wp			49.5323M	51.5017	484.045u	-7.7682n
corner1.39			ws			50.2199M	51.3273	433.752u	-4.961n
corner1.40			wz			50.1434M	50.8769	461.406u	-4.873n
corner1.41			tm			48.3844M	51.3014	472.579u	-5.1438n
corner1.42			wo			48.4412M	52.0977	466.85u	-5.8554n
corner1.43		3.4	wp			47.5763M	51.5501	497.875u	-7.7901n
corner1.44			ws			48.5943M	51.4071	446.433u	-4.9525n
corner1.45			wz			48.3083M	50.9658	474.198u	-4.8591n

PRE-LAYOUT SIMULATION (MONTE CARLO)

Monte_Carlo	avgfreq tran	avg_dutycycle tran	I_ON tran	I_OFF tranMeasure
< Filter	< Filter	< Filter	< Filter	< Filter
1	52.8228M	51.9132	474.159u	-2.7176n
2	52.1266M	51.8015	471.85u	-2.662n
3	53.5203M	51.6244	478.712u	-2.7553n
4	52.7749M	52.1618	471.739u	-2.6806n
5	53.2946M	51.9978	473.425u	-2.7178n
6	52.6811M	51.5917	473.55u	-2.7163n
7	52.5426M	51.6347	479.783u	-2.7404n
8	52.708M	51.6243	475.407u	-2.6811n
9	51.6575M	51.9386	474.539u	-2.6591n
10	53.1702M	51.9571	483.418u	-2.7223n
11	53.3613M	51.8885	475.434u	-2.7828n
12	53.6697M	51.7552	472.338u	-2.7406n
13	52.7548M	52.1994	476.235u	-2.6597n
14	52.8389M	51.9038	477.869u	-2.7435n
15	52.4283M	51.7749	460.169u	-2.7121n
16	53.102M	51.7918	483.269u	-2.7065n
17	52.4753M	51.9153	466.322u	-2.6763n
18	53.0156M	51.5344	476.085u	-2.7097n
19	53.1349M	51.9322	475.262u	-2.6889n
20	51.6666M	51.7595	480.284u	-2.7302n
21	53.4423M	52.0715	465.504u	-2.7777n
22	54.0405M	51.7721	474.261u	-2.7184n
23	52.4223M	52.1856	476.176u	-2.7441n
24	51.8509M	51.8083	469.528u	-2.7654n
25	53.1037M	52.0878	472.502u	-2.6863n
26	52.6616M	51.8608	473.353u	-2.7833n
27	53.4459M	51.7015	479.219u	-2.6247n
28	52.5419M	51.5917	468.441u	-2.7203n
29	53.9278M	52.0097	481.247u	-2.7233n
30	52.7618M	51.7578	475.549u	-2.7247n
31	51.5385M	51.7951	474.976u	-2.65n
32	52.6953M	51.6368	476.796u	-2.7328n
33	52.1532M	51.7676	471.954u	-2.7171n
34	52.7009M	51.7825	477.61u	-2.7204n
35	52.3513M	51.9145	465.93u	-2.7201n
36	55.4848M	51.7544	479.819u	-2.7464n
37	52.1527M	51.9159	479.91u	-2.6976n
38	52.903M	52.0678	477.942u	-2.7148n
39	53.2255M	51.9327	460.81u	-2.6729n
40	53.4719M	51.8215	467.623u	-2.6698n
41	52.3951M	51.7627	473.962u	-2.7572n
42	52.9559M	51.9805	469.327u	-2.6729n
43	53.2355M	52.0662	474.952u	-2.7725n
44	53.3498M	51.9656	477.475u	-2.6896n
45	51.9029M	51.9199	475.689u	-2.7077n
46	51.9688M	52.0249	467.898u	-2.6875n
47	53.3598M	52.1069	486.138u	-2.75n
48	50.671M	51.7201	469.72u	-2.7534n
49	53.543M	51.6355	476.3u	-2.6709n
50	53.5708M	51.9666	474.928u	-2.8244n
51	52.9516M	52.0216	477.85u	-2.6897n
52	52.2567M	51.7984	468.624u	-2.7163n
53	53.2044M	51.8271	480.363u	-2.6628n
54	52.6188M	51.931	475.324u	-2.7549n
55	52.8732M	51.8785	473.48u	-2.7242n
56	52.2924M	51.982	482.689u	-2.6923n
57	53.2864M	51.8737	475.854u	-2.7024n
58	53.4233M	51.6659	474.346u	-2.727n
59	50.7528M	52.0492	468.121u	-2.7078n
60	53.3669M	51.7012	485.308u	-2.6058n
61	53.4348M	51.9312	467.739u	-2.7313n
62	52.9684M	51.6895	468.838u	-2.728n
63	52.7912M	51.9479	477.046u	-2.7446n
64	52.3824M	51.8832	476.554u	-2.751n
65	52.448M	51.8168	479.558u	-2.7403n
66	52.5971M	51.8514	468.004u	-2.7146n
67	52.6354M	51.4892	480.304u	-2.7191n
68	51.88M	52.0211	467.263u	-2.7498n
69	52.5627M	51.4023	476.711u	-2.6704n
70	53.7889M	52.0343	475.173u	-2.7794n
71	53.3598M	51.9	474.512u	-2.6814n
72	53.0501M	51.5656	478.226u	-2.6723n
73	52.7043M	51.8181	470.965u	-2.8084n
74	53.552M	51.8725	476.899u	-2.6613n
75	52.4958M	51.8313	469.146u	-2.7456n
76	52.1452M	51.8361	481.984u	-2.6812n
77	53.1335M	51.5635	469.983u	-2.7318n
78	53.8064M	51.8842	474.712u	-2.6598n
79	50.3161M	51.9121	473.943u	-2.7067n
80	55.3123M	51.7038	481.662u	-2.759n
81	52.8276M	52.1651	471.107u	-2.645n
82	51.2841M	52.0617	472.781u	-2.7707n
83	53.2041M	51.7714	471.506u	-2.6959n
84	53.1988M	51.7696	475.232u	-2.6911n
85	52.3513M	51.8692	473.729u	-2.637n
86	52.2432M	51.8523	481.554u	-2.7637n
87	53.2832M	52.0697	468.202u	-2.7303n
88	51.7566M	51.8397	469.517u	-2.7344n
89	51.5047M	51.961	475.795u	-2.669n
90	52.0082M	52.1396	472.665u	-2.7309n
91	53.4239M	51.9545	473.956u	-2.7574n
92	53.0492M	52.0222	477.336u	-2.6441n
93	53.3258M	52.011	478.538u	-2.6949n
94	53.0178M	51.7517	467.829u	-2.7255n
95	53.3469M	51.8575	477.575u	-2.776n
96	52.3366M	52.1337	478.015u	-2.6423n
97	52.5972M	51.858	473.298u	-2.7511n
98	54.9091M	51.7769	478.002u	-2.7025n
99	52.3661M	51.8951	469.351u	-2.7952n
100	52.3835M	51.8522	482.075u	-2.6718n

POST-LAYOUT SIMULATION (PVT ANALYSIS)

Corner	Corner Variable temp	Corner Variable VDDA	Corner Variable xh018.lib	Corner Variable param.lib	Corner Variable config.lib	avgfreq tran	avg_dutycycle tran	I_ON tran	I_OFF tranMeasure
typ	25	3.3	tm			48.3882M	51.7372	479.149u	-3.7307n
corner1.1						51.8814M	51.9909	475.837u	-3.0556n
corner1.2			wo			51.8902M	52.491	469.634u	-3.0671n
corner1.3		3.2	wp			50.6573M	51.8265	504.533u	-2.8564n
corner1.4			ws			52.6361M	51.6771	452.145u	-3.3113n
corner1.5			wz			51.8478M	51.313	481.073u	-3.0378n
corner1.6			tm			49.5262M	52.0944	493.198u	-3.0577n
corner1.7			wo			49.5625M	52.4414	487.915u	-3.068n
corner1.8	-40	3.3	wp			48.2056M	51.9241	523.604u	-2.8521n
corner1.9			ws			50.4291M	51.9621	466.86u	-3.3071n
corner1.10			wz			49.4644M	51.4185	499.002u	-3.0274n
corner1.11			tm			47.2597M	51.9831	512.77u	-3.0398n
corner1.12			wo			47.3174M	52.543	504.852u	-3.0602n
corner1.13		3.4	wp			45.895M	51.9864	543.594u	-2.8513n
corner1.14			ws			48.2978M	51.9225	484.661u	-3.2958n
corner1.15			wz			47.1967M	51.5673	517.185u	-3.0399n
corner1.16			tm			50.4863M	51.7644	462.567u	-3.7452n
corner1.17			wo	3s	default	50.4846M	52.2773	457.121u	-3.7569n
corner1.18		3.2	wp			49.5436M	51.8702	488.427u	-3.488n
corner1.19			ws			50.8883M	51.5512	438.656u	-4.0508n
corner1.20			wz			50.4623M	51.129	466.964u	-3.7454n
corner1.21			tm			48.3882M	51.7372	479.149u	-3.7307n
corner1.22			wo			48.4123M	52.367	472.423u	-3.7507n
corner1.23	25	3.3	wp			47.3066M	51.7513	506.341u	-3.4754n
corner1.24			ws			48.9747M	51.6767	452.828u	-4.0457n
corner1.25			wz			48.3643M	51.0976	483.431u	-3.7307n
corner1.26			tm			46.379M	51.6273	495.113u	-3.7227n
corner1.27			wo			46.4054M	52.4059	488.198u	-3.7361n
corner1.28		3.4	wp			45.204M	51.9927	522.182u	-3.4814n
corner1.29			ws			47.1033M	51.8592	467.181u	-4.0261n
corner1.30			wz			46.3295M	51.3722	498.151u	-3.707n
corner1.31			tm			47.4664M	51.0961	448.427u	-6.6285n
corner1.32	150	3.2	wo			47.4249M	51.6825	444.014u	-7.2623n
corner1.33			wp			47.0077M	51.2354	474.339u	-9.3607n
corner1.34			ws			47.207M	51.1457	422.697u	-6.349n
corner1.35			wz			47.4486M	50.8186	449.592u	-6.3566n
corner1.36			tm			45.8269M	51.4607	460.311u	-6.616n
corner1.37			wo			45.826M	51.9639	455.59u	-7.2453n
corner1.38	150	3.3	wp			45.2122M	51.4014	487.207u	-9.3287n
corner1.39			ws			45.8033M	51.2878	435.229u	-6.366n
corner1.40			wz			45.7936M	50.8003	463.766u	-6.3664n
corner1.41			tm			44.2242M	51.402	474.393u	-6.6132n
corner1.42			wo			44.1983M	52.0369	469.053u	-7.1925n
corner1.43		3.4	wp			43.4535M	51.5824	500.876u	-9.3473n
corner1.44			ws			44.4196M	51.4254	447.687u	-6.3436n
corner1.45			wz			44.1793M	50.7797	478.431u	-6.3645n

My Parameters

Parameters	Minimum	Typical	Maximum	% Difference
Average Frequency (MHz)	43.5692	48.5132	52.066	4.42
Average Duty Cycle (%)	50.6634	51.7024	52.6052	0.4
I_ON(uA)	422.639	478.305	542.624	0.3

Required Specs:

Parameters	Minimum	Typical	Maximum
Average Frequency (MHz)	45.689	50.7581	55.149
Average Duty Cycle (%)	50.426	51.5007	52.284
I_ON(uA)	426.17	479.6	542.47
I_OFF(nA)	1.9962	2.7691	7.9005

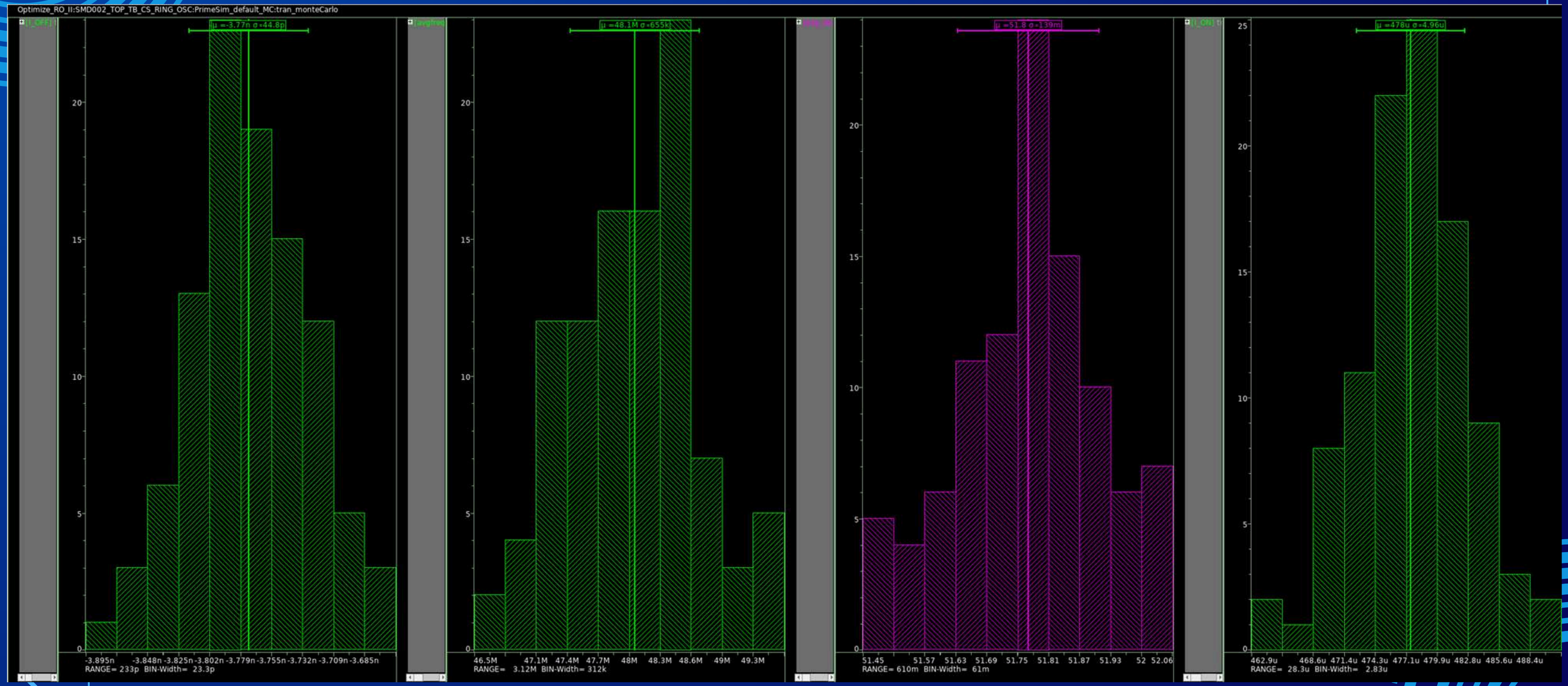
POST-LAYOUT SIMULATION (MONTE CARLO)

- Monte Carlo is a statistical simulation that predicts how random manufacturing defects will affect the final performance of your fabricated chips.
- In a real fabrication plant, no two transistors or wafer lots are printed perfectly identically. Small variations constantly occur in oxide thickness, doping concentration and physical dimensions (known as local mismatch or process variation).
- The simulator (Custom Compiler) uses the random variable in the PDK file provided by foundry and then running 100 simulations.
- After that, we calculate the min value and standard deviation of average frequency and duty cycle(%).

POST-LAYOUT SIMULATION (MONTE CARLO)

Monte_Carlo	avgfreq tran	avg_dutycycle tran	I_ON tran	I_OFF tranMeasure
< Filter	< Filter	< Filter	< Filter	< Filter
1	48.0969M	51.8607	476.989u	-3.7753n
2	47.6104M	51.8003	475.254u	-3.7199n
3	48.565M	51.8731	478.789u	-3.8178n
4	48.8765M	51.9758	480.842u	-3.7605n
5	47.753M	51.8915	471.461u	-3.7747n
6	47.785M	51.6053	473.377u	-3.7672n
7	47.9315M	51.6857	483.315u	-3.8017n
8	48.4668M	51.4467	481.471u	-3.7457n
9	47.337M	51.6335	479.561u	-3.7106n
10	47.6105M	51.8924	480.333u	-3.7885n
11	48.675M	51.7629	478.113u	-3.8278n
12	48.3354M	51.8273	474.395u	-3.8052n
13	47.4197M	51.8194	475.729u	-3.7077n
14	47.6693M	51.8005	477.643u	-3.7894n
15	49.1854M	51.9159	472.64u	-3.7926n
16	47.3231M	51.6362	478.847u	-3.7684n
17	48.478M	51.8016	474.691u	-3.725n
18	48.5688M	51.9098	481.16u	-3.764n
19	48.2787M	51.502	477.351u	-3.7706n
20	47.8339M	51.691	488.635u	-3.7781n
21	47.7601M	51.9361	462.939u	-3.855n
22	48.8879M	51.7158	475.904u	-3.7816n
23	48.3447M	51.8365	483.262u	-3.8158n
24	47.8735M	51.8209	477.709u	-3.8226n
25	48.4145M	51.8464	474.86u	-3.7368n
26	48.4175M	51.6777	478.375u	-3.8364n
27	48.1655M	51.6837	479.949u	-3.6622n
28	47.5748M	51.9165	469.075u	-3.7827n
29	47.9311M	51.567	478.072u	-3.7987n
30	47.6774M	51.9691	476.01u	-3.7867n
31	47.8879M	51.5508	486.396u	-3.6857n
32	48.3616M	51.6649	483.194u	-3.8009n
33	48.3952M	51.6958	479.091u	-3.7685n
34	47.2028M	51.7604	474.883u	-3.7681n
35	47.9316M	51.7052	471.201u	-3.7827n
36	49.0661M	51.8194	473.314u	-3.7927n
37	47.9704M	51.8283	484.431u	-3.7469n
38	47.2892M	51.889	475.601u	-3.7639n
39	49.2985M	51.7755	469.777u	-3.7548n
40	48.194M	51.9754	466.783u	-3.7197n
41	48.6243M	51.7807	481.595u	-3.8225n
42	47.7457M	51.7914	469.916u	-3.7115n
43	48.5056M	51.8908	479.291u	-3.844n
44	47.8161M	51.5435	477.21u	-3.7276n
45	48.3038M	51.5953	487.667u	-3.7607n
46	47.943M	51.8458	475.537u	-3.7425n
47	48.2588M	51.6854	488.08u	-3.8155n
48	47.0628M	51.8644	476.877u	-3.8235n
49	48.9271M	51.699	480.084u	-3.7384n
50	48.2184M	52.0564	472.494u	-3.895n
51	48.2167M	51.7934	481.281u	-3.749n
52	46.6541M	51.7649	465.184u	-3.7734n
53	48.5998M	51.7556	483.922u	-3.7278n
54	47.9049M	51.9009	478.751u	-3.8248n
55	48.2373M	51.8122	475.495u	-3.7491n
56	46.8567M	51.4636	481.873u	-3.73n
57	49.1577M	51.7733	481.936u	-3.7643n
58	47.7027M	51.7818	472.453u	-3.7915n
59	47.3371M	51.7061	478.655u	-3.763n
60	47.2013M	51.6918	481.013u	-3.6955n
61	49.5321M	51.8036	476.933u	-3.773n
62	48.6401M	51.6903	473.757u	-3.7898n
63	47.6521M	51.9975	477.022u	-3.7895n
64	46.934M	51.6613	475.343u	-3.8125n
65	47.268M	51.9985	479.817u	-3.7788n
66	48.4792M	51.8709	474.163u	-3.7813n
67	47.8293M	51.448	481.476u	-3.7722n
68	48.0616M	51.7342	474.65u	-3.7802n
69	48.2964M	51.513	483.364u	-3.7462n
70	48.4768M	52.0084	474.57u	-3.8419n
71	48.5854M	51.8564	478.213u	-3.7374n
72	47.1494M	51.9487	472.755u	-3.7497n
73	47.575M	51.6272	470.483u	-3.8696n
74	48.3235M	51.6043	476.963u	-3.7075n
75	48.7809M	51.7428	478.677u	-3.7882n
76	48.3841M	51.6164	491.261u	-3.7281n
77	48.224M	51.7026	469.989u	-3.7914n
78	48.5278M	51.9563	474.677u	-3.7215n
79	46.4553M	51.7909	483.008u	-3.7511n
80	49.5747M	51.6915	480.284u	-3.8445n
81	48.4086M	51.7743	478.377u	-3.7202n
82	47.0361M	52.0556	478.051u	-3.8326n
83	48.0897M	52.0125	472.482u	-3.7825n
84	48.5061M	51.5773	478.347u	-3.7782n
85	48.0634M	51.7835	478.885u	-3.6752n
86	47.1212M	51.8605	480.396u	-3.8139n
87	49.2888M	51.7685	476.516u	-3.7901n
88	47.9127M	51.7631	477.27u	-3.7918n
89	47.5404M	51.8146	484.285u	-3.7337n
90	47.3912M	51.7321	477.395u	-3.785n
91	48.2585M	51.7938	474.71u	-3.8189n
92	48.4444M	51.9992	481.243u	-3.7025n
93	47.2014M	51.7914	473.51u	-3.744n
94	48.5327M	51.7551	470.548u	-3.7737n
95	48.788M	51.4874	484.364u	-3.818n
96	47.5476M	51.9115	481.068u	-3.6751n
97	48.4136M	51.7722	477.727u	-3.8142n
98	49.5764M	51.6555	478.991u	-3.7488n
99	47.4833M	51.8955	470.597u	-3.8519n
100	47.124M	51.671	481.038u	-3.7179n

POST-LAYOUT SIMULATION (MONTE CARLO)



CALCULATIONS – MONTE CARLO

Range of value for 3 sigma calculation :

Parameters	Mean	Standard Deviation	Range (3 sigmas)
Average Frequency (Hz)	48.1	655k	[46.14M,50.07M]
Average Duty Cycle (%)	51.8	0.139	[51.38,52.22]
I_ON(uA)	478	4.96	[463.1,492.9]
I_OFF(A)	−3.77n	44.8p	[−3.904ns,−3.636ns]

$$Range = \mu \pm 3\sigma$$

CALCULATIONS – COEFFICIENT OF VARIATION (CV)

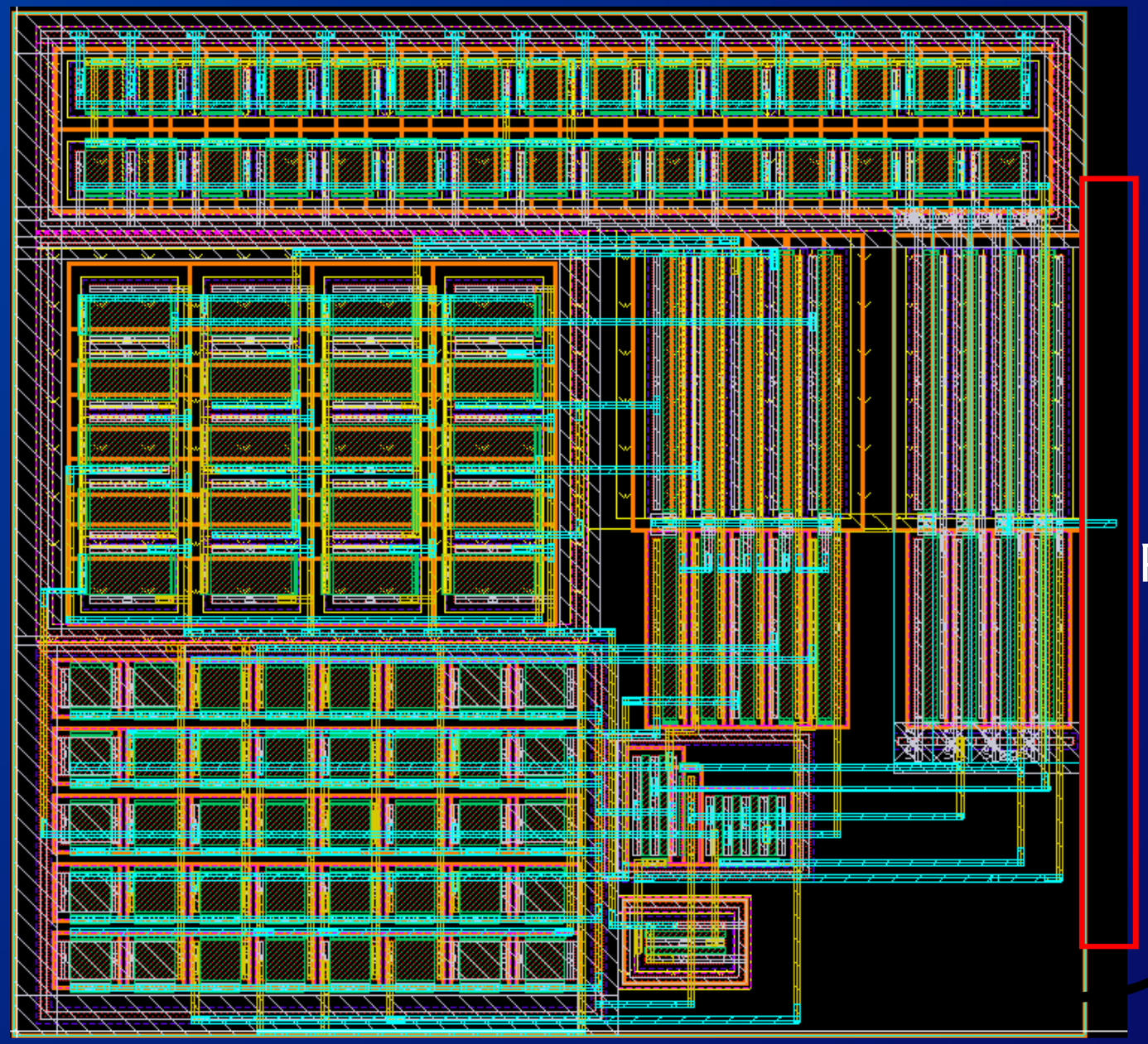
The Coefficient of Variation measures the relative variation of the parameter:

$$CV(\%) = \frac{\sigma}{\mu} \times 100$$

Parameters	Mean	Standard Deviation	CV
Average Frequency (Hz)	48.1	655k	$CV = \frac{0.655}{48.1} \times 100\%$ $CV = 1.36\%$
Average Duty Cycle (%)	51.8	0.139	$CV = \frac{0.139}{51.8} \times 100\%$ $CV = 0.27\%$
I_ON(uA)	478	4.96	$CV = \frac{4.96}{478} \times 100\%$ $CV = 1.04\%$
I_OFF(A)	−3.77n	44.8p	$CV = \frac{0.0448}{3.77} \times 100\%$ $CV = 1.19\%$

Usually the CV value of < 5% considered good enough and < 2% to be very stable oscillator

FINAL LAYOUT



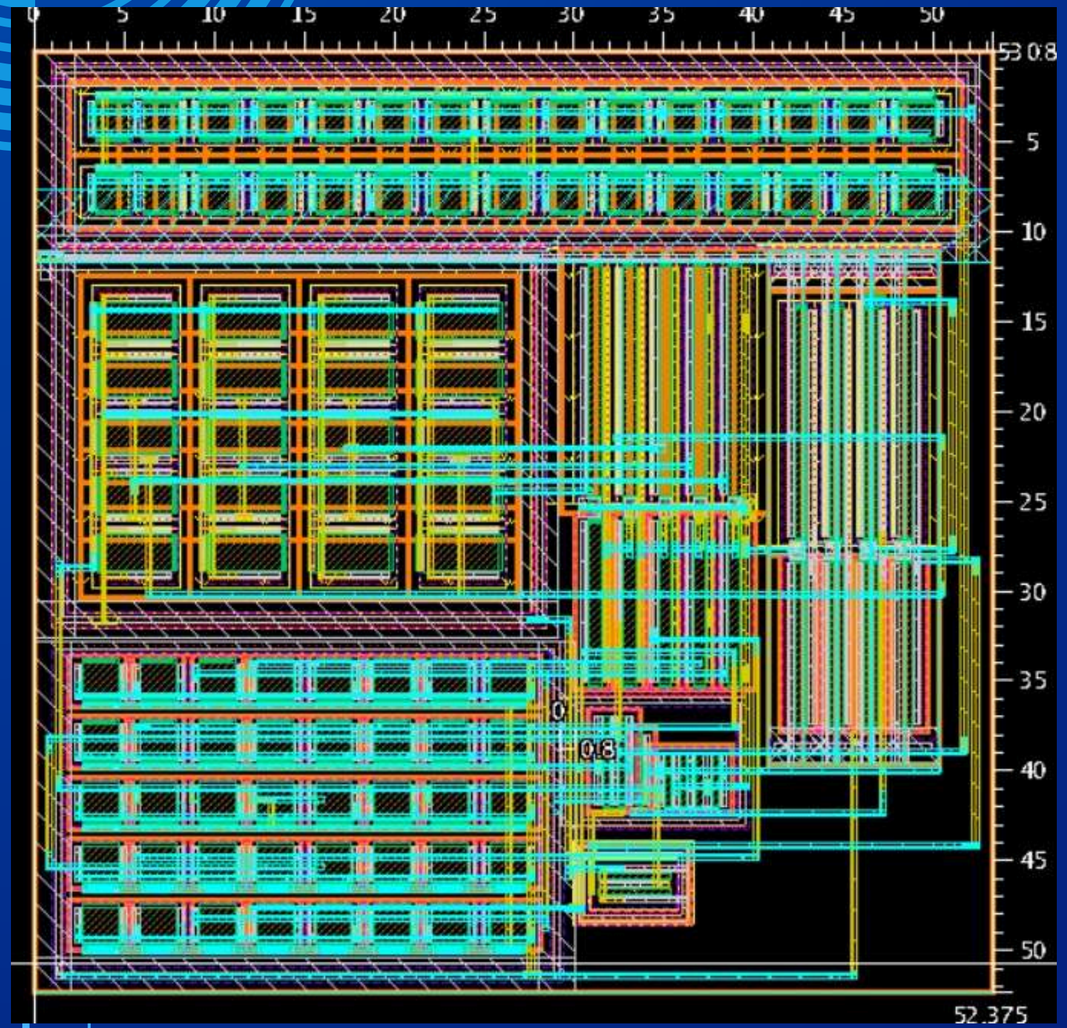
Area efficient

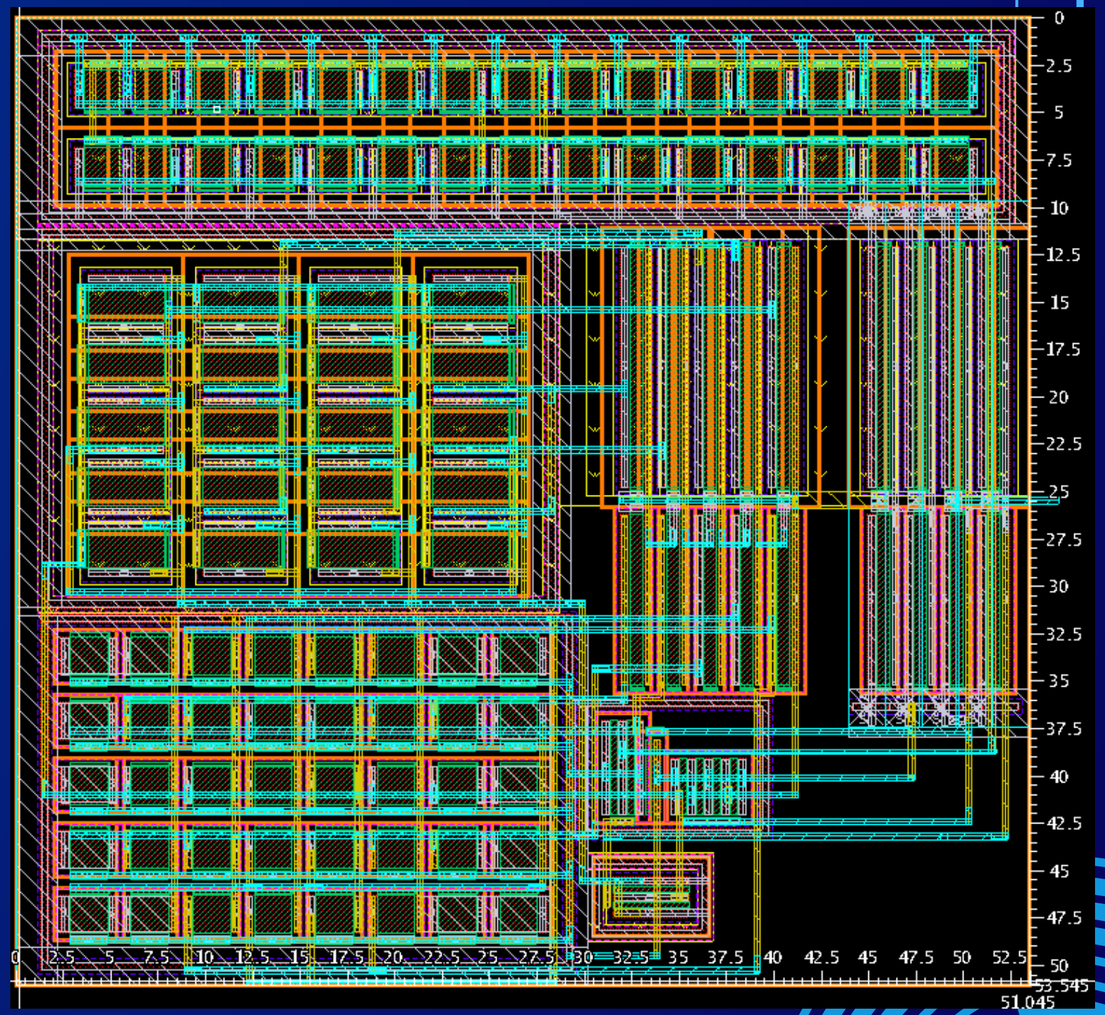
Tidier Routing

Pin placement

Exploiting deadspace

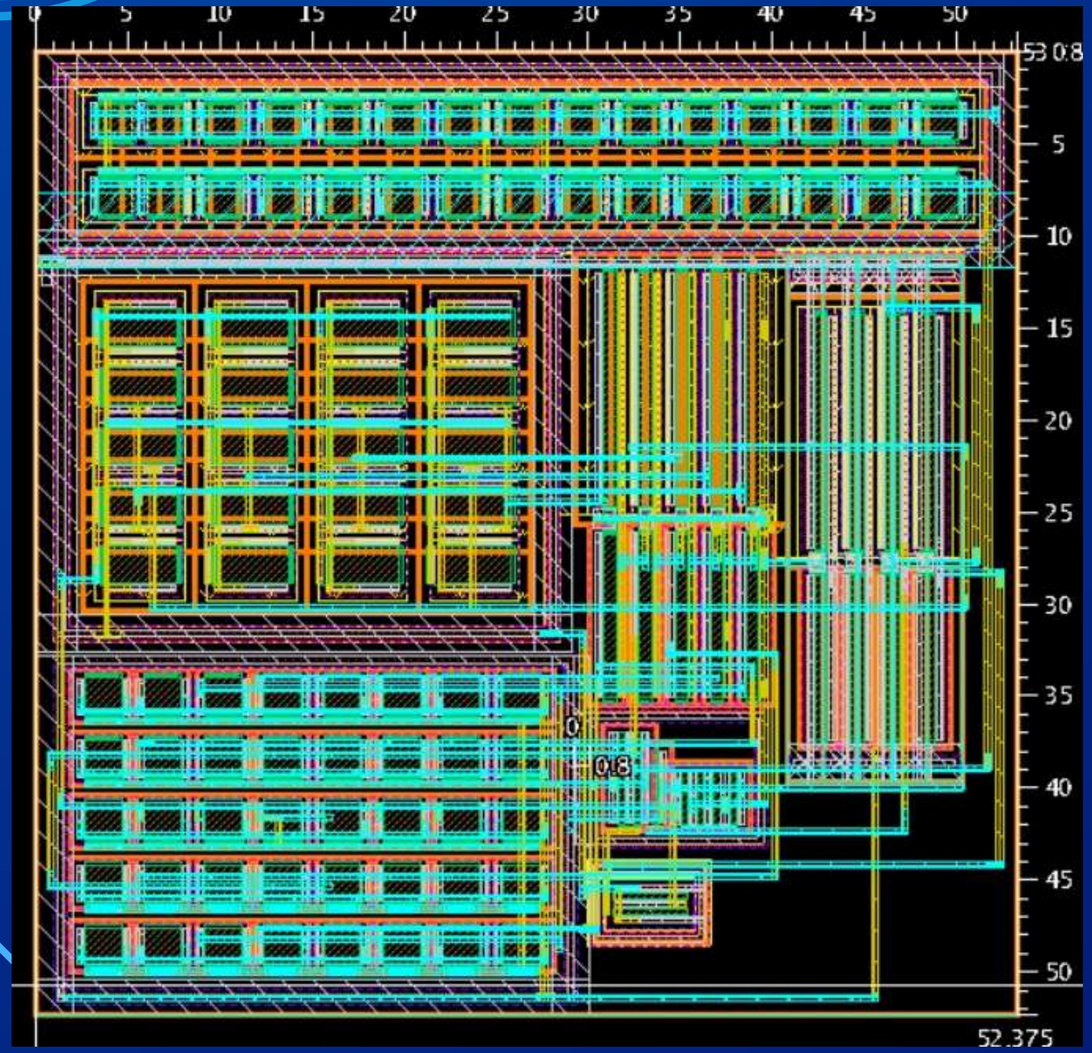
IMPROVEMENT





IMPROVEMENT

PREVIOUSLY...



Disadvantanges:

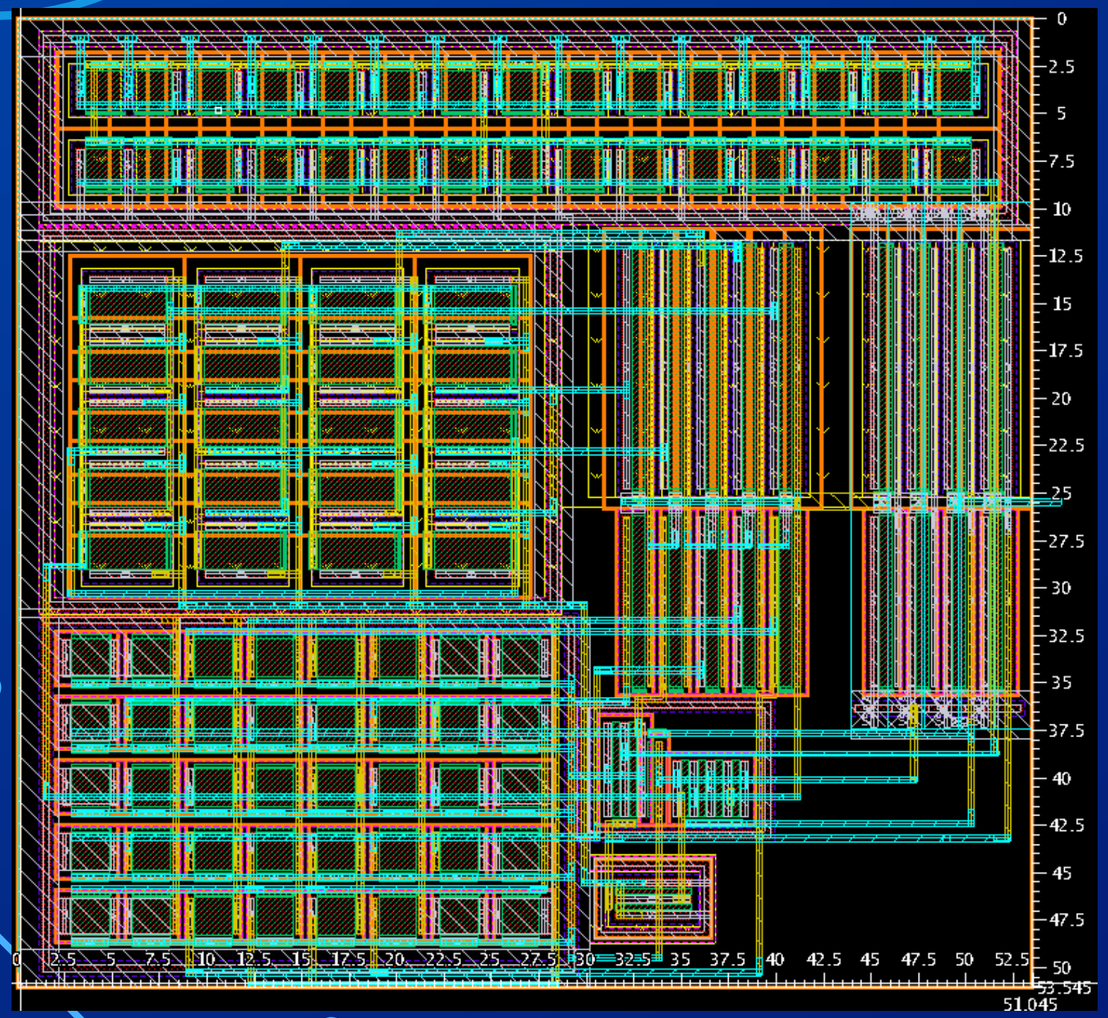
- Longer metal routing
- Many metal routing on device
- Quite messy

Effects:

- This can produce higher parasitic values
- Hence, this will lead the circuit to be more sluggish.
- Routing on device can introduce mechanical stress to the transistors nearby.

IMPROVEMENT

CURRENTLY...



Advantanges:

- Shorten metal routing
- Minimizing metal routing on device
- More organized and structred. Tidier routing.
- Distance away the inverter from current mirror to minimize cross-talk/noise.

Effects:

- This lowers parasitic values
- Hence, this will lead to faster ring oscillator manufactured.
- More enhanced PVT results.

CHALLENGES & SOLUTIONS

The Problem 1: The parasitic problem

- The major problem is to reduce the parasitic. One way is using higher metal, shorter metal routing and reduce on-body routing.

Problem 2: The routing methodology

- Because we are aiming for compact layout, the room for routing quite minimal. We need to scratch our head to find for the space to route.

Problem 3: Determining the good placement/floorplan

- The first step we encounter was the hardest one since we need to determine the best floorplan before beginning our routing step.

SUMMARY

- We successfully designed and implemented the layout design of a 50MHz Ring Oscillator and running the post-layout simulation successfully.
- Post-layout extraction confirmed a typical oscillation frequency of 48.51 MHz and a highly balanced 51.7% duty cycle which is close to the theoretical typical value within 4% and 0.4% difference respectively.
- We implement layout matching techniques(Common Centroid), compact floorplan and using dummy transistors in order to mitigate process gradients and edge effects.
- We obey the DRC and LVS rules as well and the efforts of reducing the parasitic of the circuits.

Parameters	Minimum	Typical	Maximum	% Difference
Average Frequency (MHz)	43.5692	48.5132	52.066	4.42
Average Duty Cycle (%)	50.6634	51.7024	52.6052	0.4
I_ON(uA)	422.639	478.305	542.624	0.3

THANK YOU